The galaxy-bias ladder across thin redshift bins

A correlated linear-bias prior, one-loop bias models, and the accuracy of DESI C_ℓ^{gg}

November 2026

1 Summary

The question. Thin redshift bins measure growth only if the galaxy bias of each bin is tied to its neighbors. This note sets how freely the linear bias of a thin bin may move within its tracer group, tests whether each choice still recovers the cosmology honestly, and gives the figure of merit (FoM) of the choices that pass. The data are DESI angular band powers C_ℓ^{gg} alone, on four bin layouts (Section 2).

The prior. Each bin's bias follows its group's growth-tied trend B_g/D times a free straight line in redshift, plus a residual with a correlated Gaussian prior of width σ_g and correlation length L (Section 3). Four priors span the range: growth-tied (no line, no residual), baseline ($\sigma_g = 3\%$ for BGS and 1% elsewhere, $L = 0.2$), pessimistic (3% everywhere, $L = 0.05$) and free (one free bias per bin).

Model A. The higher-order biases follow each bin's linear bias through calibrations, with one offset, one counterterm and one excess white noise per group (model A baseline); model A pessimistic also tilts the offsets and the counterterm in redshift. The scale cut is $k_{\rm max} = 0.075\,h/{\rm Mpc}$ at $z = 0$ (Section 4).

The accuracy test. Every prior fits 2000 random bias histories: a drift derived from physics plus a residual drawn from the baseline or the pessimistic prior (Section 5). The growth-tied prior fails on every layout and is discarded. The baseline prior is honest only when the truth is as smooth as it assumes: under the pessimistic residuals its Ω_m error bar covers the truth in only 40–63% of universes instead of 68% on three of the four layouts. The pessimistic and free priors are honest under both. A linear drift of the quadratic and tidal bias calibrations moves (w_0, w_a) of model A baseline by more than a third of its error on the wide layouts; model A pessimistic absorbs it by construction.

The headline. Model A pessimistic on the pessimistic prior is the forecast that stays honest under every truth tested (Figure 5). Model A baseline on the baseline prior is what better knowledge of bias evolution and of the calibrations would buy, and the free prior is the lower limit. On every prior, DESI C_ℓ^{gg} alone sits well below LSST Y1 3x2pt in both figures of merit at this cut.

Limits. The shifts are linear responses, the truths are theory rather than mocks, and model A has real-space loops with linear redshift-space distortions (Appendix R). Appendix S lists the open issues: the one-loop terms left out of the covariance, the scale cut built on the linear model, negative counterterms in chains, and the third-order bias template.

2 The forecast

The data are the DESI angular band powers C_ℓ^{gg} of thin redshift bins, computed without the Limber approximation, for six tracer groups (BGS, LRG1, LRG2, LRG3, ELG1 and ELG2) on four bin layouts: bins of equal comoving width $\Delta d_{\rm com} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. Bin i has mean redshift $\bar{z}_i$, and $D(z)$ is the linear growth factor, normalized to $D(0) = 1$. The covariance is Gaussian, with the linear bias $B_g/D(\bar{z}_i)$ of each bin.

$w(a) = w_0 + w_a(1 - a)$ is the dark-energy equation of state, with a the scale factor; Ω_m is the matter density today in units of the critical density, and A_s the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$. A parameter's marginalized one-sigma error is σ. The figure of merit $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) error ellipse, and $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ is defined the same way.

The scale cut. $k_{\rm max}$ labels one cut per bin. Bin i is cut at $k_{\rm max}/D(\bar{z}_i)$: matter is less nonlinear at higher redshift, so the cut moves to higher wavenumber k as $1/D$. The cut is the non-Limber UV-fraction cut of Saraivanov et al. [1], which follows DeRose and Chen [2]. Each band power is split into $C_\ell^{\rm IR}$, computed from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it, and a band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \; < \; 0.05. \tag{1}$$

The kept bands are intersected with the shipped mask. The cut was built on the linear-bias model. Every $k_{\rm max}$ in this note is the $z = 0$ value.

The linear-bias run. The notebook `EXAMPLE_EVALUATE_FISHER.ipynb`, called v1 in the notebooks, runs the same ladder with linear bias only; this note calls it the linear-bias run. It built the mechanics, and model A runs on its kept band powers, so every difference from it is the bias model.

3 The linear-bias prior

The default model ties all bins of a group to one amplitude, $b_{1,i} = B_g/D(\bar{z}_i)$, with $b_{1,i}$ the linear bias of bin i and B_g one free amplitude per group g. This growth-tied model keeps the clustering amplitude $b_1 D$ of a group constant, as measured for DESI-like luminous red galaxies [14]. The tying is what makes thin bins powerful, and it fails in two ways. It is too rigid: known physics moves each bin away from the group trend at the percent level (Appendix B). And it is inaccurate: a drift the model cannot express displaces the fitted cosmology by several times its own error (Section 5). One free bias per bin is accurate by construction but spends part of

the data measuring the biases, 76 of them on $\Delta d_{\rm com} = 62$ Mpc. The prior below sits between the two: bins may deviate from the group trend, but neighboring bins deviate together.

The linear bias of bin i in group g is

$$b_{1,i} \;=\; \frac{B_g}{D(\bar{z}_i)}\,(1+\delta_i), \qquad \delta_i \;=\; s_g\,(\bar{z}_i - \bar{z}_g) + \varepsilon_i, \tag{2}$$

with B_g a free amplitude and s_g a free tilt per group (flat priors), $\bar{z}_g$ the mean of the group's bin redshifts, δ_i the fractional deviation from the group trend, and ε_i the residual around the group's straight line. The residuals carry a Gaussian prior correlated within a group and zero across groups,

$$\langle \varepsilon_i\, \varepsilon_j \rangle \;=\; \sigma_g^2\, \exp\!\left(-\,\frac{|\bar{z}_i - \bar{z}_j|}{L}\right), \tag{3}$$

with σ_g the prior width of one bin's residual in group g and L the correlation length in redshift. The tilt is free for two reasons: the shape B_g/D is not the evolution of a conserved population, whose bias obeys $b - 1 \propto 1/D$ [9] (Appendix B.1), and every derived drift is close to a straight line across its group (Appendix C). In the Fisher matrix the prior becomes one projected precision block per group that charges only the residual, never the level or the line (Appendix C.5). A group of one or two bins is therefore free, since a level and a line fix every bin. Every group of the $\Delta z = 0.2$ layout is of this kind, so on that layout the baseline, pessimistic and free priors coincide.

What σ_g and L mean. σ_g sets how far one bin may wander from its group's line. L sets how far apart two bins must be before their residuals are independent: their correlation is $e^{-\Delta z/L}$ at a separation Δz. Together they set the rms jump of the residual ε between neighbors, $\sigma_g\sqrt{2\,(1 - e^{-\Delta z/L})}$ (Appendix A.2). Physically, the line absorbs the coherent evolution of a selected sample, such as a flux threshold that rises with distance or the debiasing of a conserved population. The residual holds what a line cannot: the curvature of such a drift, and local features of the selection (Appendix C.4).

The four priors.

- **Growth-tied**: $b_{1,i} = B_g/D(\bar{z}_i)$, with no tilt and no residual; the default of the shipped yamls.
- **The baseline prior**: $\sigma_g = 3\%$ for BGS, 1% for the other five groups, and $L = 0.2$. The BGS width holds the curvature of its flux-limit drift. The $r < 19.5$ magnitude limit becomes a luminosity threshold that rises with redshift and moves the bias 22% away from the growth-tied shape across the group; a straight line removes all of that except a 2.4% rms curvature (Appendix E). The 1% of the other groups is a stated hedge for selection curvature not modeled here, at least six times the residual a conserved population leaves after its line (at most 0.16% rms, Appendix C.4). $L = 0.2$ is comparable to the group spans, so the residual is smooth across a group.
- **The pessimistic prior**: $\sigma_g = 3\%$ in every group and $L = 0.05$. It places local selection features in every group, as large as the largest derived curvature, on the redshift scale over which the 4000 Å break crosses a filter edge. That crossing is known, but its amplitude in bias is not measured (Appendix B.3), so this prior grants it the largest derived size everywhere.

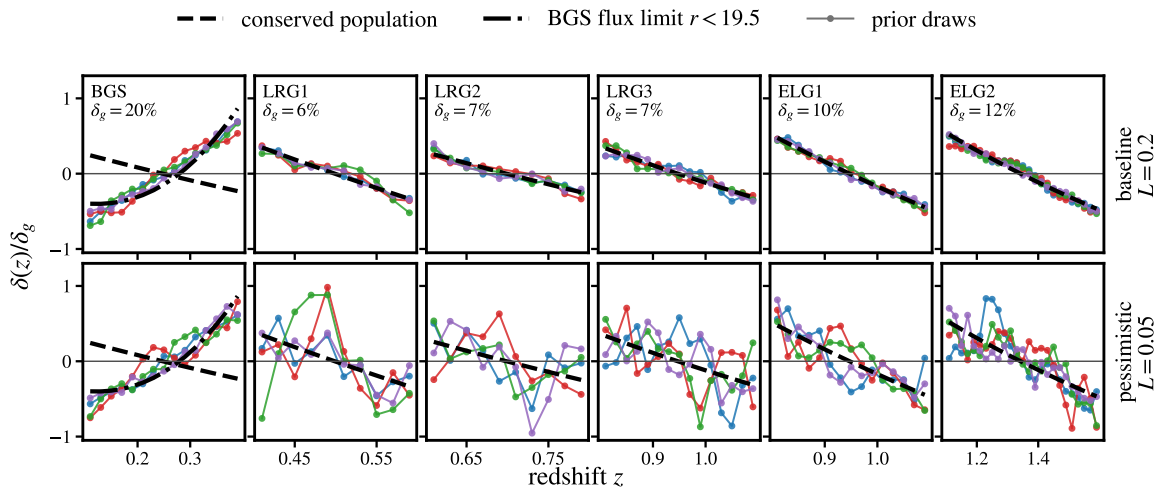


Figure 1: Draws of the two stated priors on the $\Delta z = 0.02$ bins, one column per group: the baseline prior (top row; $\sigma_g = 3\%$ for BGS and 1% elsewhere, $L = 0.2$) and the pessimistic prior (bottom row; $\sigma_g = 3\%$, $L = 0.05$). Each column is divided by the round constant δ_g in its top panel, so all panels share one axis. Thick dashed: the conserved-population drift; thick dash-dot (BGS): the $r < 19.5$ flux-limit drift. Thin curves: four draws of $\delta(z)$, each on the line of the group's reference drift (the flux limit in BGS, the conserved population elsewhere), so draw and drift differ by the residual ε.

- **Free**: one $b_{1,i}$ per bin, each with a flat prior; the lower limit.

Figure 1 draws both stated priors around the known drifts. Under the baseline prior the neighbor rms of the residual at $\Delta z = 0.02$ is 0.44% (1.31% for BGS), the draws are smooth, and the known drifts sit well inside the family: the curvature the BGS line leaves costs a prior penalty of about 0.8 per charged direction on the $\Delta d_{\rm com} = 62$ Mpc bins. Under the pessimistic prior the neighbor rms is 2.44% in every group: the histories keep their lines but gain local features nobody has measured, which is the point of that prior.

4 Model A

Linear bias alone is not enough at this cut: in the companion note `notes/linear_bias_kmax`, a linear-bias fit to a one-loop truth at $k_{\rm max} = 0.075\,h/\mathrm{Mpc}$ moves Ω_m by 1.5–1.8σ and (w_0, w_a) by 0.28–0.55σ on the $\Delta d_{\rm com} = 62$ Mpc layout under the four linear-bias priors, and linear bias is safe only at $k_{\rm max} = 0.05\,h/\mathrm{Mpc}$. Model A, of `notes/higher_order_bias`, keeps the per-bin linear bias of Eq. (2) and adds the one-loop terms. As computed here, the real-space galaxy power spectrum of bin i is

$$P_{gg,i}(k) \;=\; b_{1,i}^2\,P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_{1,i}, b_{2,i}, b_{s^2,i}) \;+\; \alpha_0\,k^2 P_{\rm mm}(k) \;+\; \frac{\mathrm{SN}_0}{\bar{n}_i}, \tag{4}$$

where $P_{\rm mm}$ is the nonlinear matter power spectrum and $P_{\rm loop}$ holds the one-loop terms. These carry the quadratic bias b_2, the response of the galaxy density to the square of the matter

density contrast δ_m (the galaxy density contrast contains $b_2\,\delta_m^2/2$), and the tidal bias b_{s^2}, its response to the squared tidal field; the third-order bias b_3 is zero. α_0 is the k^2 counterterm, in $h^{-2}\,\mathrm{Mpc}^2$, which absorbs the sensitivity of the loops to small scales. SN_0 is the white-noise power beyond Poisson, in units of the bin's Poisson power $1/\bar{n}_i$, with $\bar{n}_i$ the bin's number density. The higher-order biases follow each bin's own linear bias through calibrations, plus one offset per group:

$$b_{2,i} \;=\; b_2^{\mathrm{L}}(b_{1,i}) + \Delta_{2,g}, \qquad b_{s^2,i} \;=\; -\tfrac{4}{7}\,(b_{1,i}-1) + \Delta_{s,g}, \tag{5}$$

with b_2^{L} the separate-universe fit of Lazeyras et al. [3] and the second relation the co-evolution of a population formed with local bias [4].

The priors are Gaussian and centered on zero: widths 1 on Δ_2, 0.5 on Δ_s, 2 on SN_0 and $0.5/k_{\mathrm{max}}^2 = 88.9\,h^{-2}\,\mathrm{Mpc}^2$ on α_0. The α_0 width is the rule of the DESI 2024 full-shape analysis [5, 6] moved to this cut: $\alpha_0 k^2 = 0.5$ at $k = k_{\mathrm{max}}$, a 50% correction relative to P_{mm} at the $z = 0$ cut. At bin i's own cut, $k_{\mathrm{max}}/D(\bar{z}_i)$, the same width gives $\alpha_0 k^2 = 0.5/D(\bar{z}_i)^2$. Relative to the linear galaxy power $b_{1,i}^2 P_{\mathrm{mm}}$ both are smaller by a further factor $b_{1,i}^2$. Redshift-space distortions are linear (Kaiser), with each bin's redshift distribution convolved by the sampled velocity dispersion σ_v (the Fingers-of-God smearing), as in the linear-bias run.

Two variants of the higher-order sector sit on every linear-bias prior. **Model A baseline** has one Δ_2, Δ_s, α_0 and SN_0 per group, 24 parameters. **Model A pessimistic** also tilts the Δ_2, Δ_s and α_0 of each group linearly in redshift, with flat priors on the tilts, up to 42 parameters. SN_0 is not tilted, by choice: its Poisson part is known bin by bin, and the excess is kept at one number per group. Appendix M describes the implementation and the Fisher matrices.

5 The accuracy test

No mock catalogs and only two derived drift laws are available, so this note does not defend one prior. It tests each prior against random truths and quotes the FoM only for the priors that pass.

Random truths. Each truth sets every bin's linear bias to its group's known drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population elsewhere; Appendix F.2) plus a residual drawn from the baseline or the pessimistic prior, 2000 truths per family (Appendix F.3). The truth's b_2 and b_{s^2} follow its b_1 through Eq. (5). Every linear-bias prior fits every truth under both variants, centered on the growth-tied shape, to first order in the truth (Appendix F.1). Figure 2 draws the (w_0, w_a) shift of each fit in units of its own ellipse, without data noise. Where a fit's prior equals the truth's scatter, the mean shift is not a bias: that fit's ellipse already includes this scatter (Appendix F.4).

The statistic. Each fit is scored by

$$\sqrt{\langle\chi^2\rangle/n_{\mathrm{dof}}}, \qquad \chi^2 = \Delta p^{\mathsf{T}}\,\mathsf{C}_p^{-1}\,\Delta p, \tag{6}$$

with Δp the recovered minus the true parameters, C_p the fit's marginalized covariance, $n_{\mathrm{dof}} = 2$ for $p = (w_0, w_a)$ and 1 for Ω_m or A_s, and $\langle\cdot\rangle$ the exact average over the data noise and the bias history (the known drift plus a random residual). Its value, written η, is 1 for error bars that

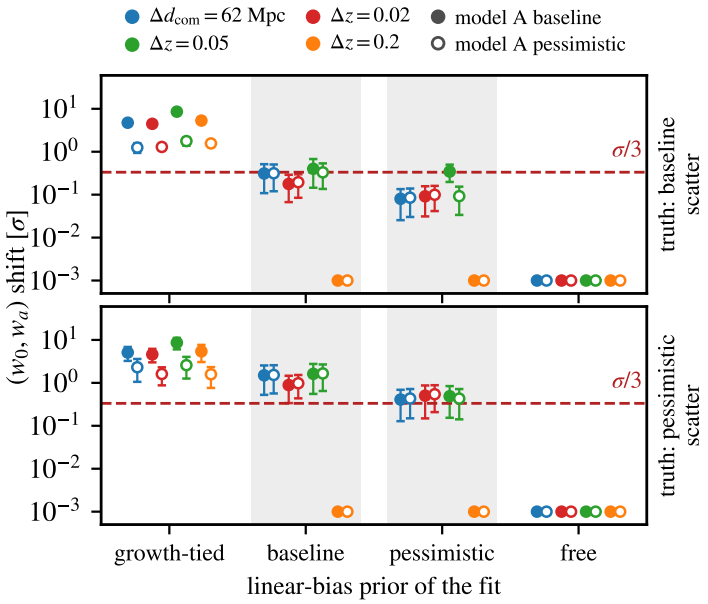


Figure 2: The (w_0, w_a) shift of each fit in units of its own marginalized ellipse, under model A, for 2000 truths: the known drifts plus a residual drawn from the baseline prior (top) or the pessimistic prior (bottom), without data noise. Filled: model A baseline; open: model A pessimistic; colors: the layouts. Points: the mean shift; bars: the 16th–84th percentiles. Gray columns: the baseline and pessimistic priors. Dashed line: $\sigma/3$. Points at 10^{-3}: exact zeros. $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

cover the truth as often as they claim; above 1 they are too small, and below 1 conservative. It scores the recovery of the parameters, not the χ^2 of the data: a biased fit can match the band powers while its cosmology is wrong. Appendix F derives it and converts it into the fraction of universes in which a one-parameter 1σ bar covers the truth; the percentages below come from that conversion.

The result of the test. Figures 2 and 3 order the priors.

- *The growth-tied prior fails under every truth and is discarded.* The conserved-population drift alone moves its (w_0, w_a) by several σ (Appendix N), and its error bars are far too small under both scatters; under the pessimistic scatter its Ω_m bar covers the truth in only 5–39% of universes. Under model A pessimistic, whose flat-prior tilts follow part of a linear-bias drift, η falls to 1.4–2.3 in (w_0, w_a), still far from honest. A prior whose error bars do not cover the truth has no meaningful FoM.

- *The baseline prior is honest only if the bias residual is as smooth as it assumes.* Under its own scatter η is 1.00–1.02. Under the pessimistic scatter it fails on every layout but $\Delta z = 0.2$ (Section 3): the Ω_m bar covers the truth in only 56, 63 and 40% of universes on $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$ and 0.05, and the A_s bar in 57, 65 and 54%. Because the noise term of $\langle\chi^2\rangle$ is at most one, the rms displacement beyond the noise is at least $(\eta^2 - 1)^{1/2}$ error bars (Appendix F.4). Model A pessimistic does not change this.

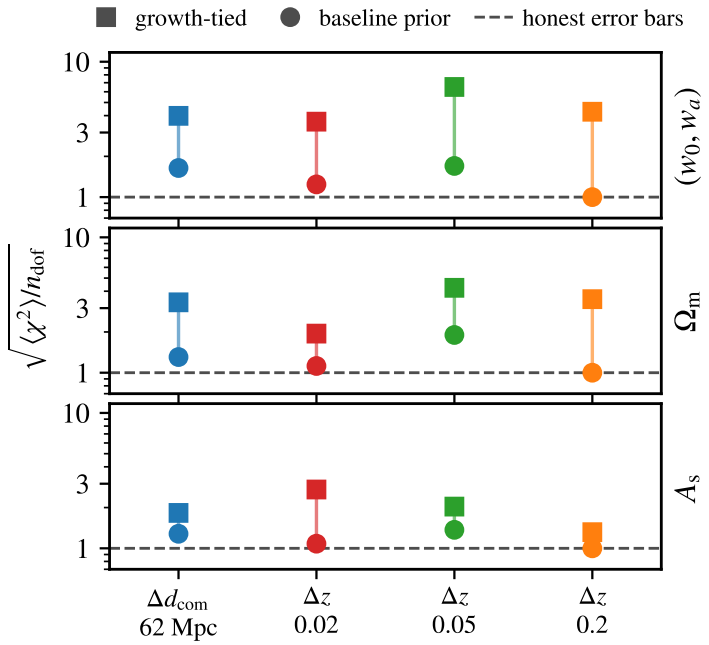


Figure 3: How well each fit's quoted error bar covers where the fit actually lands, measured by $\eta = \sqrt{\langle\chi^2\rangle/n_{\rm dof}}$: $\chi^2 = \Delta p^{\mathsf{T}} \mathsf{C}_p^{-1} \Delta p$, with Δp the recovered minus the true parameters and C_p the fit's marginalized covariance; $n_{\rm dof} = 2$ for (w_0, w_a), 1 for Ω_m or A_s; $\langle\cdot\rangle$ averages over the data noise and the bias history (the known drifts plus a residual from the pessimistic prior). For one parameter, the 1σ bar covers the truth in a fraction $2\Phi(1/\eta) - 1$ of the universes (Φ the Gaussian cumulative distribution): 68% at $\eta = 1$ (dashed line), 38% at $\eta = 2$. Model A baseline; squares: the growth-tied prior; circles: the baseline prior. Omitted: the free prior (exactly 1) and the pessimistic prior (at most 1.05). $k_{\rm max} = 0.075\,h/{\rm Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

- *The pessimistic prior is honest under both scatters*: $\eta \leq 1.05$ in every score, and 0.89–1.00 in (w_0, w_a) (conservative) under baseline scatter. The 1.05 comes from the BGS flux-limit drift on $\Delta z = 0.05$, which this prior leaves just above $\sigma/3$ (Appendix N). The free prior gives exactly 1.

A drift of the calibrations. The random truths move only the linear bias; the higher-order biases follow it through calibrations assumed exact up to one constant offset per group. A third truth tests that assumption. It drifts the offsets linearly with the bin redshift across each group, Δ_2 from about -0.5 to $+0.5$ and Δ_s from about -0.25 to $+0.25$ (half the prior width at each end), and keeps b_1 at its fiducial (Appendix M.4). Under model A baseline (Figure 4) the shift of (w_0, w_a) exceeds $\sigma/3$ under every prior on the two wide layouts, $\Delta z = 0.05$ and 0.2, and the A_s shift of the baseline, pessimistic and free priors reaches $\sigma/3$ on $\Delta z = 0.05$. On the two thin layouts only the growth-tied prior exceeds $\sigma/3$. Model A pessimistic absorbs this drift exactly, by construction: its tilts have the drift's shape. The test shows only that the tilts work for a linear drift; a nonlinear drift of the calibrations within a group is untested.

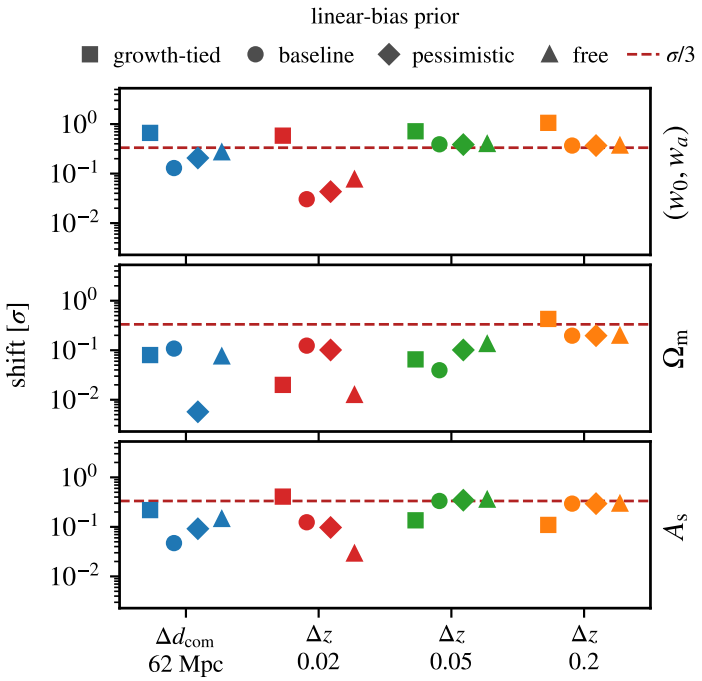
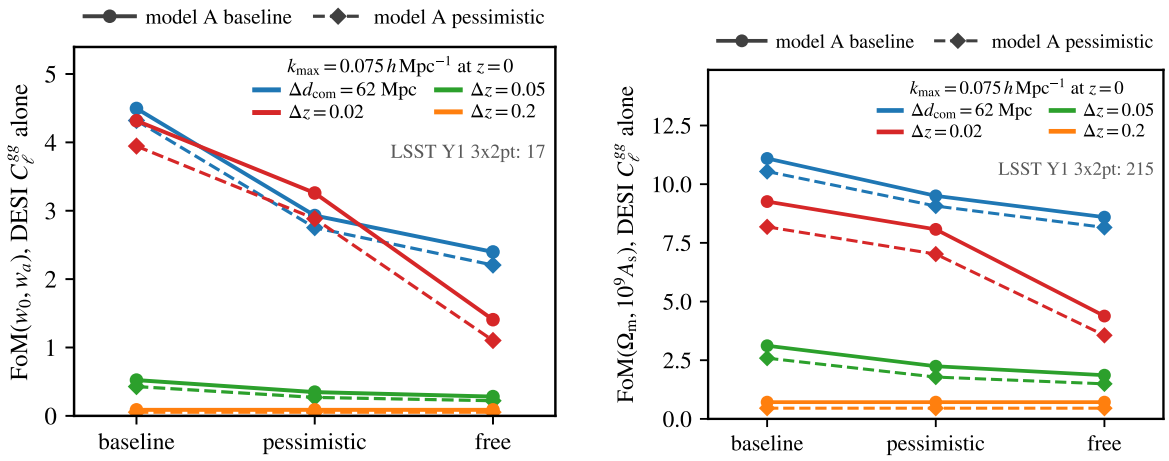


Figure 4: The shift of a model A baseline fit under the calibration drift (Δ_2 and Δ_s changing linearly by 1 and 0.5 across each group): (w_0, w_a) in units of the fit's ellipse, Ω_m and A_s in their own σ, per layout. Markers: the four linear-bias priors. Dashed line: $\sigma/3$. Omitted: model A pessimistic, which absorbs the drift exactly by construction. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

6 The figure of merit

A FoM means something only for a prior that passes the accuracy test: the pessimistic and free priors under both truths, the baseline prior only if the bias residual is as smooth as it assumes. Figure 5 therefore draws these three priors, with FoM(w_0, w_a) in panel (a) and FoM$(\Omega_m, 10^9 A_s)$ in panel (b), both from the same Fisher inverse. Both fall from the baseline prior to free under each variant, except on $\Delta z = 0.2$, where the three priors coincide. The two thin layouts carry most of the information. Model A pessimistic costs little against model A baseline on the thin layouts and more on the wide ones. DESI C_ℓ^{gg} alone sits well below LSST Y1 3x2pt [7] in both figures of merit at this cut. Appendix Q sets these FoMs next to the linear-bias run and to exact calibrations, neither of which is a result here.

Result. The headline is model A pessimistic on the pessimistic prior: it is honest under both families of random truths and absorbs a linear calibration drift exactly. Model A baseline on the baseline prior shows what better knowledge of bias evolution would buy. It holds only under two assumptions: each bin's linear-bias residual is no larger and no rougher than the baseline prior, and the calibrations of b_2 and b_{s^2} hold within a group up to one constant offset. If the first fails at the pessimistic size, the Ω_m bar covers the truth in only 40–63% of universes; if the second fails by the linear drift above, (w_0, w_a) moves by more than $\sigma/3$ on the



(a) (b)

Figure 5: **(a)** FoM(w_0, w_a) and **(b)** FoM$(\Omega_m, 10^9 A_s)$ of DESI C_ℓ^{gg} alone under the baseline, pessimistic and free priors. Colors: the bin layouts. Solid: model A baseline; dashed: model A pessimistic. The text in each panel quotes the LSST Y1 3x2pt value of the same figure of merit, which lies far above the DESI points. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

wide layouts. The free prior under model A pessimistic is the lower limit, and the growth-tied prior is discarded.

A The models

This appendix first builds the prior without the tilt of Eq. (2): the kernel model, in which the exponential kernel describes the whole deviation δ_i of a bin from its group's trend. Appendix C adds the free tilt, and the kernel then describes the residual ε_i around each group's line. In the appendices b_i is the linear bias $b_{1,i}$ of bin i.

A.1 Growth-tied, free, and correlated biases

The growth-tied model sets $b_i = B_g/D(\bar{z}_i)$, with one free amplitude B_g per group. The kernel model multiplies it by a fractional deviation δ_i,

$$b_i \;=\; \frac{B_g}{D(\bar{z}_i)}\,\big(1+\delta_i\big), \tag{7}$$

with B_g still free (flat prior). The deviations receive a Gaussian prior that is correlated inside a group and zero across groups,

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_g^2\,\exp\!\left(-\,\frac{|\bar{z}_i-\bar{z}_j|}{L}\right). \tag{8}$$

In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{9}$$

so that $b_i = \beta_i/D(\bar{z}_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the β_i within one group is

$$\mathsf{C} \;=\; \sigma_\beta^2\,\mathsf{R}, \qquad \sigma_\beta \;=\; \sigma_g\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|\bar{z}_i-\bar{z}_j|}{L}\right), \tag{10}$$

with σ_β the spread of one amplitude, in units of a bias, and R the dimensionless correlation matrix, with ones on the diagonal.

A.2 What σ_g and L mean

The diagonal of the kernel gives the wander of a single bin, $\mathrm{Var}(\delta_i) = \sigma_g^2$: σ_g sets how far any one bin may drift from the group's trend. The correlation coefficient of two bins follows from the same kernel,

$$r \;=\; \frac{\langle \delta_i \delta_j \rangle}{\sqrt{\langle \delta_i^2 \rangle\,\langle \delta_j^2 \rangle}} \;=\; e^{-\Delta z/L}, \tag{11}$$

with $\Delta z = |\bar{z}_i - \bar{z}_j|$ the bin separation: σ_g cancels, so the correlation is set by the separation and L alone. The jump between the two bins has variance $\mathrm{Var}(\delta_i) + \mathrm{Var}(\delta_j) - 2\,\mathrm{Cov}(\delta_i, \delta_j)$,

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_g^2\left(1 - e^{-\Delta z/L}\right) \;\approx\; \frac{2\,\sigma_g^2\,\Delta z}{L} \quad (\Delta z \ll L). \tag{12}$$

So L sets the correlation, σ_g the wander, and together they set the jump a thin bin may take relative to its neighbor. At small separations the jump variance grows linearly with Δz, as the distance walked by a random walker grows linearly in time: below L the deviation curve $\delta(z)$ looks like a random walk, and only beyond L does the finite wander pull it back.

Worked example. Neighbors at $\Delta z = 0.02$ with $L = 0.05$ have $\Delta z/L = 0.4$ and $r = e^{-0.4} = 0.670$. The exact rms jump of Eq. (12) is $\sigma_g\sqrt{2\,(1-0.670)} = 0.81\,\sigma_g$, and the small-separation form gives $\sigma_g\sqrt{0.8} = 0.89\,\sigma_g$. The approximation overestimates because $x = \Delta z/L = 0.4$ is not small: x exceeds $1 - e^{-x}$ already at second order ($x^2/2$), by about twenty percent here, and the square root halves that.

A.3 The free group amplitude and the projection

A plain Gaussian prior with covariance C would penalize moving all β_i of a group up or down together, but that uniform shift is exactly the free amplitude B_g, which the growth-tied model never constrains. The repair removes the uniform direction from the prior: with $\mathbf{1}$ the ones vector of the group's n bins, the prior precision of a group's amplitudes is the projection

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{13}$$

Four statements carry the construction. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing. A pure shape change keeps exactly the penalty the plain prior gives it. P is positive semidefinite with rank $n-1$: one direction is free and the other $n-1$ are charged. And P is precisely the prior obtained by sampling B_g with a flat prior and marginalizing it. Appendix G computes P for two bins, Appendix H derives it by marginalization, and Appendix I treats n bins. Across groups the prior is zero, so the full prior precision is block diagonal, one projected block per group (Appendix K).

B Which L and σ_g are physical

One case of bias evolution can be solved in closed form: a population whose members are conserved. This appendix derives that case, measures its deviation from the growth-tied trend, lists what breaks conservation in a selected sample, and weighs the published measurements.

B.1 A conserved population: the bias continuity equation

Take a sample whose membership never changes, assume the galaxies move with the matter on large scales (no velocity bias), and work in linear theory. The matter overdensity δ_m and the galaxy-count overdensity $\delta_{\rm gal}$ then obey the same linear continuity equation with the same peculiar velocity field, so the excess clumping $\delta_{\rm gal} - \delta_m$ is frozen (Appendix D spells out every step). With b_c the linear bias ($\delta_{\rm gal} = b\,\delta_m$) at a reference redshift z_c and $\delta_m \propto D$, carrying the frozen difference to any other redshift gives [9]

$$b(z) - 1 \;=\; \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}. \tag{14}$$

The excess clumping set when the galaxies formed stays fixed while the matter clustering grows underneath it, so the bias decays toward one with time. Tegmark and Peebles [10] find the same decay for stochastic bias (Appendix D).

The growth-tied shape $b = B/D$ freezes a different quantity: constant $b\,D$ means $\delta_{\rm gal} = b\,\delta_m \propto B$, a galaxy clustering amplitude that does not grow. A conserved population never does that. Constant $b\,D$ instead describes a selection that picks, at each redshift, a different and rarer set of objects, whose stronger clumping compensates the smaller D.

B.2 The deviation from the growth-tied trend

Place the reference at a group's midpoint z_c, with $b_c = B/D(z_c)$ from the group's fiducial amplitude B. The ratio of the conserved bias, Eq. (14), to the growth-tied bias $b_c\, D(z_c)/D(z)$ defines the deviation $\delta(z)$ of Eq. (7):

$$1 + \delta(z) \;=\; \frac{D(z)/D(z_c) + b_c - 1}{b_c}, \qquad \delta(z) \;=\; \frac{D(z)/D(z_c) - 1}{b_c}. \tag{15}$$

At the midpoint $\delta = 0$; at lower redshift the matter has grown, so $\delta > 0$, and at higher redshift $\delta < 0$. Because $D(z)$ is smooth, $\delta(z)$ is nearly linear across a group.

In flat ΛCDM with $\Omega_m = 0.3$ and the project's fiducial amplitudes, Eq. (15) gives $\delta = +5.3\%$ and -4.9% at the ends of BGS ($B = 1.34$, $0.1 < z < 0.4$, $b_c = 1.5$), $+2.4\%$ and -2.2% at the ends of LRG1 ($B = 1.70$, $0.4 < z < 0.6$, $b_c = 2.2$), and $+2.6\%$ and -2.3% at the ends of LRG3 ($B = 1.70$, $0.8 < z < 1.1$, $b_c = 2.7$). Between neighbors the conserved deviation changes by 0.3–0.7% at $\Delta z = 0.02$ and by 0.8–1.7% at $\Delta z = 0.05$. In the language of the prior it is a smooth tilt: coherent over the whole group, so L longer than the group width. For comparison, the neighbor rms of Eq. (12) at $\Delta z = 0.02$ (and 0.05) is 2.4% (3.4%) for $\sigma_g = 3\%$ and $L = 0.05$, and 0.44% (0.67%) for $\sigma_g = 1\%$ and $L = 0.2$.

Figure 6 draws these shapes. In every group the dashed conserved curve is flatter than the solid B_g/D, because the frozen excess clumping is divided by a growing D. On BGS the dash-dot flux-limit curve is the opposite: much steeper than B_g/D, and convex.

B.3 What breaks conservation in a selected sample

Threshold drift. A target selection holds a roughly constant comoving density until the flux limit bites; past that redshift the sample keeps only its brighter members, which sit in more massive halos and are more biased. The bias change is smooth, with L comparable to the group width.

Color-selection features. Selection cuts in color space can respond quickly in redshift when a spectral feature crosses a filter edge. The sharpest common feature is the 4000 Å break: a filter edge 100–200 Å wide maps to $\Delta z \approx 0.03$–0.05 at that break, and photometric scatter smears the transition further. The crossing is known; its amplitude in bias is unmeasured.

Mergers. Two members that merge become one, so the number of galaxies is not conserved even before any selection acts.

Emission-line selection. An ELG enters the sample while its emission lines are strong, so the membership follows the star-formation phase of each galaxy rather than one conserved set of objects.

B.4 What the measurements say

Each statement below carries a tag: *measured* (read from the cited source), *derived* (computed here from measured inputs) or *assumed* (a judgment not backed by a measurement). Equation (14) is from Fry [9], which has no arXiv record; Tegmark and Peebles [10] (section "After the galaxy formation epoch") rederive it.

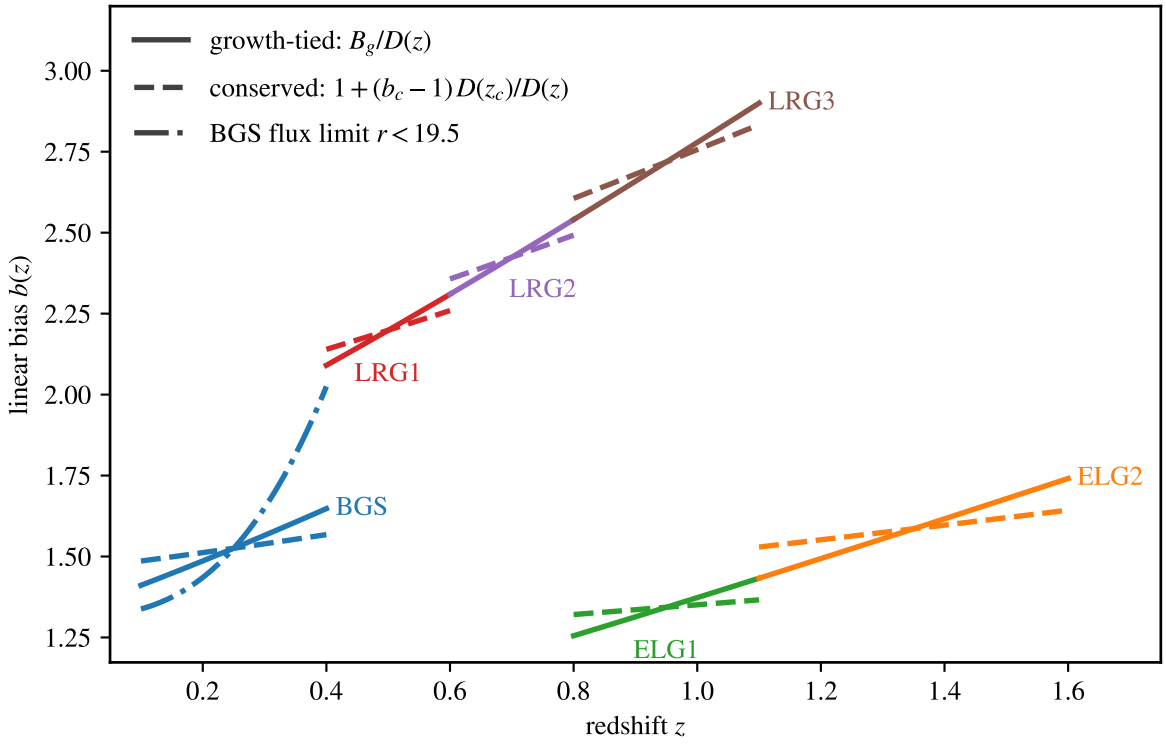


Figure 6: The bias $b(z)$ of every group. Solid: the growth-tied shape $B_g/D(z)$. Dashed: a conserved population anchored at the group midpoint (Eq. (14)). Dash-dot, on BGS only: the $r < 19.5$ flux-limit bias, anchored at the same midpoint.

Massive red galaxies follow the conserved shape.

- Wake et al. [11]: SDSS and 2SLAQ LRGs at $0.19 < z < 0.55$, matched in color and luminosity, show no significant evolution of the clustering amplitude and a merger rate of $7.5 \pm 2.3\%$ over that range, about 2.4% per Gyr (measured).
- Tojeiro et al. [12]: BOSS progenitors of the SDSS LRGs, weighted by their fossil record, merge at $2 \pm 1.5\%$ per Gyr and cluster as the continuity equation predicts (measured).
- Guo et al. [13]: CMASS samples of fixed comoving density at $0.43 < z < 0.7$ cluster like a conserved population under gravity (measured).
- Zhou et al. [14]: the bias of DESI-like LRGs in photometric bins of $\Delta z \approx 0.1$ stays within $\pm 2.3\%$ of the constant clustering amplitude $b = 1.5/D$ below $z = 0.83$, with the highest bin $6.4 \pm 6.4\%$ high (measured biases, residuals derived; Figure 7).

The selections drift.

- Zhou et al. [15]: the LRG cuts hold the comoving density near $5 \times 10^{-4}\,(h/\mathrm{Mpc})^3$ at $0.4 < z < 0.8$ and hand over to a fixed flux limit beyond (measured). That limit raises the luminosity threshold with the distance modulus by 0.27 magnitudes per $\Delta z = 0.1$

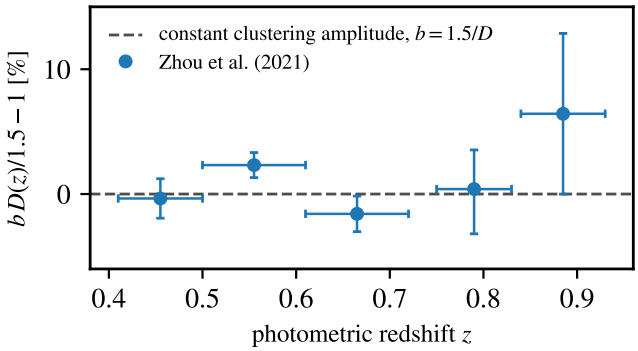


Figure 7: The measured bias of DESI-like LRGs [14] as a fractional residual around the constant clustering amplitude $b = 1.5/D$, with D at each photometric bin's center (flat ΛCDM, $\Omega_m = 0.3$); horizontal bars: the bin edges.

at $z \approx 1$; times the threshold-bias slope below, it drifts the LRG3 bias by 6–11% per $\Delta z = 0.1$ on top of the B/D shape (derived, from an r-band relation used outside its calibrated range, so only the direction and order of magnitude are established).

- Zehavi et al. [20]: the large-scale bias of SDSS luminosity-threshold samples at $z < 0.25$ is $b\,(\sigma_8/0.8) = 1.06 + 0.21\,(L_{\rm th}/L_*)^{1.12}$, a slope of 0.5–1.0 in bias per magnitude near LRG luminosities (measured). It converts a threshold drift into a bias drift.

- Hahn et al. [23]: BGS Bright is magnitude limited at $r < 19.5$ (measured), which drives a 10–25% bias drift across the BGS group (derived; Appendix E).

The ELG bias shape is stable.

- Rocher et al. [25]: HOD fits of the DESI One-Percent ELGs in $0.8 < z < 1.1$ and $1.1 < z < 1.6$ agree within one sigma (measured); at these host masses the halo bias depends weakly on mass, so the implied drift is at the percent level (derived, slope assumed).

- Raichoor et al. [24]: the [OII] doublet that carries the ELG redshift loses signal-to-noise in narrow windows, $\Delta z \approx 0.001$–0.01 wide, where it lands on sky emission lines, especially at $z > 1.5$ (measured). This is the one short-separation mechanism with a measured location; its amplitude in bias is assumed, of order 1–2%, from the weak dependence of ELG bias on [OII] flux.

Nothing resolves bias below $\Delta z \approx 0.1$. The lensing tomography of the same samples, White et al. [17] and Sailer et al. [18, 19] on the photometric samples of Zhou et al. [16], uses bins of about 0.15 with one free bias per bin and finds smooth models that fit (measured context). Below $\Delta z \approx 0.1$ the thin-bin statements of this note rest on the continuity equation and on the selection arguments.

C The memory model

C.1 Why a second model

Every deviation found above, except the sky-line windows, is a monotonic drift: the conserved-population tilt (about 1% per $\Delta z = 0.05$), the LRG3 magnitude-limit drift (6–11% per $\Delta z = 0.1$) and the BGS flux-limit drift (10–25% across the group). A monotonic drift is a coherent, long-correlation deformation, not bin-to-bin decorrelation. The kernel model can admit such a drift only at a price: to pass a 20% drift with $L \leq 0.1$ it takes $\sigma_g = 10$–15%, and that width also allows roughly 10% scatter between neighbors, which no identified mechanism produces. One σ_g controls both the drift and the scatter, so the kernel cannot say "large smooth drift, small local scatter". The memory model separates the two.

C.2 Why the kernel forgets

The kernel's correlation $e^{-|\bar{z}_i - \bar{z}_j|/L}$ decays with separation: at $L = 0.05$, bins 0.15 apart share $e^{-3} \approx 0.05$ of their deviation. With $r_k = e^{-(\bar{z}_{k+1} - \bar{z}_k)/L}$ the correlation of the neighbors k and $k+1$, the best guess of the next bin given all earlier ones is (Appendix I.2)

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; r_k\, \delta_k : \tag{16}$$

only the nearest neighbor matters, and the influence of any one bin fades as the product of the neighbor correlations along the chain.

C.3 The model, and why it remembers

The memory model adds one number per group, a tilt, and splits the deviation into a line and a residual,

$$\delta_i \;=\; s_g\, v_i + \varepsilon_i, \qquad v_i \;=\; \bar{z}_i - \bar{z}_g, \tag{17}$$

with $\bar{z}_g$ the mean of the group's bin redshifts, s_g a free tilt with a flat prior (the same status as B_g), and the residual ε_i carrying the kernel of Eq. (8). This is the model of Eqs. (2) and (3).

To see what the tilt changes, give the slope a Gaussian prior of variance τ^2 first (the flat prior is the limit $\tau \to \infty$). The deviations then have covariance

$$\mathrm{Cov}(\delta_i, \delta_j) \;=\; \tau^2\, v_i\, v_j \;+\; \sigma_g^2\, e^{-|\bar{z}_i - \bar{z}_j|/L}. \tag{18}$$

The new term does not decay with separation: bins on the same side of the group center are positively correlated, bins on opposite sides negatively. Far beyond L the kernel term is gone, and the correlation coefficient tends to

$$\rho_{ij} \;\longrightarrow\; \frac{\tau^2 v_i v_j}{\sqrt{\big(\tau^2 v_i^2 + \sigma_g^2\big)\big(\tau^2 v_j^2 + \sigma_g^2\big)}} \;\longrightarrow\; \pm 1 \quad \text{when } \tau|v| \gg \sigma_g : \tag{19}$$

the two ends of a group remember each other through the shared slope. Prediction shows the same memory. The best guess of the next bin becomes

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; \hat{c} + \hat{s}_g\, v_{k+1} \;+\; r_k\, \big(\delta_k - \hat{c} - \hat{s}_g\, v_k\big), \tag{20}$$

where $(\hat{c}, \hat{s}_g)$ are the level and slope of the best straight line through all the past bins, weighted by C^{-1} (at least two past bins). Every past bin enters through the fitted line; only the wiggle around the line stays Markov. Statistics calls this the universal-kriging predictor; Appendix J proves it.

Figure 8 shows the difference on the BGS bins at $\Delta z = 0.02$. Kernel draws at $\sigma_g = 10\%$, about what the kernel requires to admit a drift of BGS size, zigzag and forget, and none looks like the flux-limit drift. Memory draws with a tilt plus $\sigma_g = 2\%$ are lines with small wiggles, and the drift looks like a typical draw.

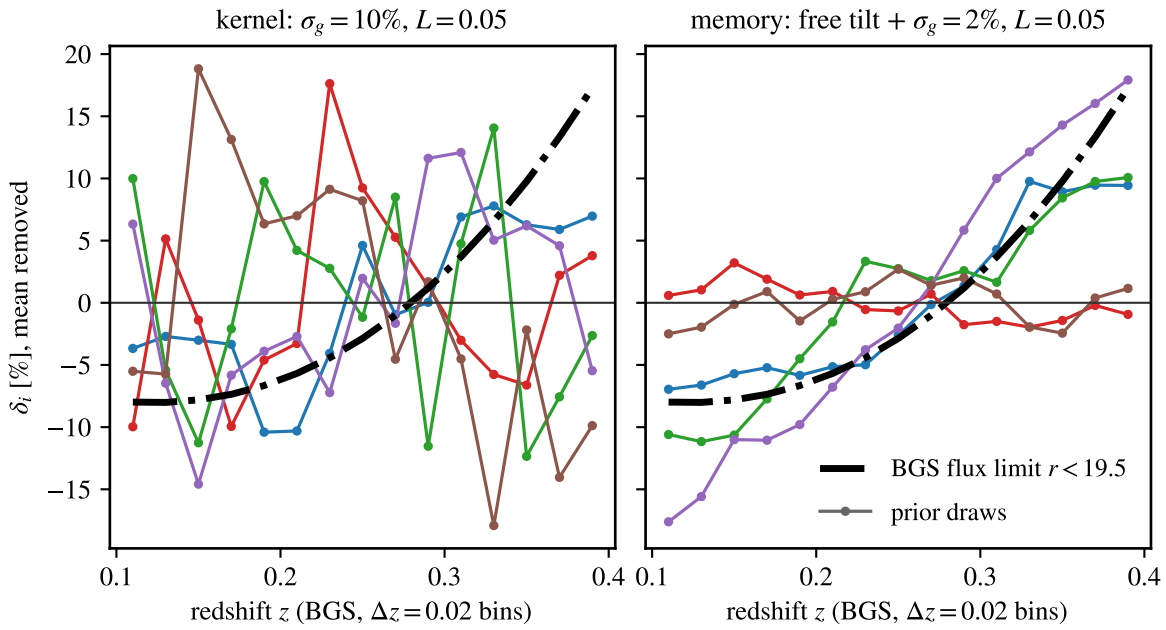


Figure 8: Random prior draws of the deviations of the 15 BGS bins at $\Delta z = 0.02$, each draw's mean removed (B_g is free). Thick dash-dot: the BGS flux-limit drift. Left: the kernel model. Right: the memory model, whose tilt is free in the model and drawn here, for display only, from a Gaussian of width 0.6 per unit redshift.

C.4 Why the memory model differs physically

A selected sample changes along a group in two ways. Coherent changes move every bin the same way: a flux or magnitude threshold that rises with distance, the debiasing of a conserved population, the drift of the halo-mass threshold at fixed number density. Local changes are features of the selection that move bins independently and are forgotten over a short range, such as the sky-line windows and filter-edge crossings of Appendix B.3. The memory model gives the first kind to the tilt and the second to the kernel.

One straight line is enough for the coherent kind, because the drifts are smooth functions of redshift, set by $D(z)$ and the distance modulus, which change on Δz of order one. The best equal-weight line through each group's conserved deviation at the $\Delta z = 0.05$ bin centers (derived; flat ΛCDM with $\Omega_m = 0.3$, amplitudes 1.34 for BGS, 1.70 for the LRGs and 0.84

for the ELGs) removes deviations that span 3–11% across a group and leaves a residual of 0.02–0.16% rms, at most 0.27% (ELG2).

The BGS flux-limit drift is larger and convex: it spans 22.3%, and its best line leaves 2.4% rms and 3.3% at most (Appendix E). For BGS the residual width of the memory model is therefore about 2–3%, the curvature a line cannot absorb. Figure 9 shows both statements: every conserved deviation is a straight line, while the BGS flux limit leaves a U shape that the residual must absorb.

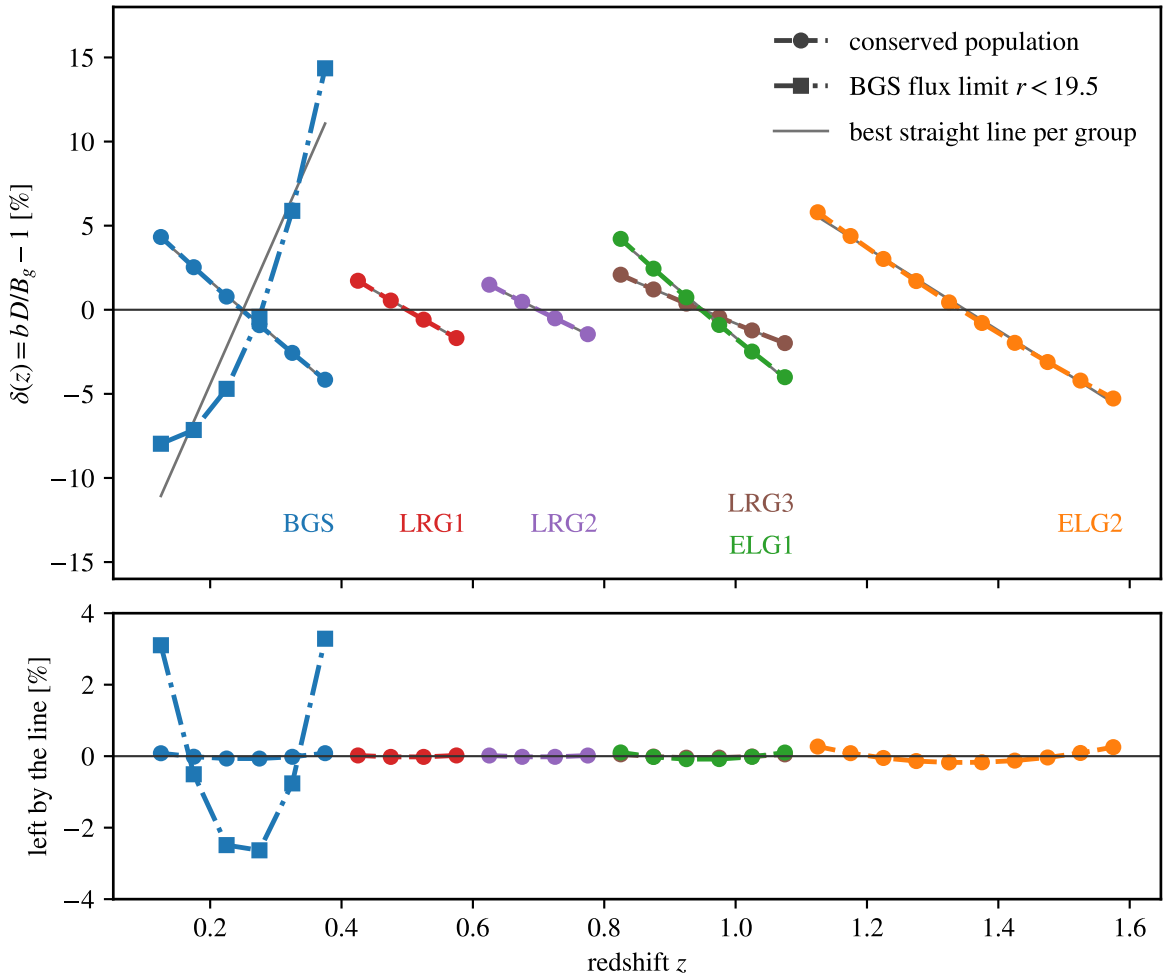


Figure 9: Top: the deviation $\delta(z) = b\,D/B_g - 1$ at the $\Delta z = 0.05$ bin centers, with B_g set so the mean deviation vanishes. Dashed with circles: the conserved population, every group. Dash-dot with squares: the BGS flux limit. Thin gray: the best straight line per curve. Bottom: what the line leaves.

The two models give σ_g different physical meanings. In the kernel model it is the total departure of a bin from the B_g/D trend, drift plus scatter; in the memory model it is the departure from the group's own line, curvature plus local scatter. The plausible values differ accordingly: about 0.1% for a conserved population, 1–2% for the local selection features (Appendix B.4), and 2–3% for the BGS curvature. The memory model rests on the same

evidence as the kernel model; it encodes that evidence in a form closer to the physics.

C.5 What the prior charges

Give the line's level a Gaussian prior of width W and its slope one of width τ; the flat priors of the model are the limit $W, \tau \to \infty$. In that limit the prior precision of a group's bin amplitudes becomes (Appendix J)

$$\mathsf{P}_{\text{mem}} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \qquad \mathsf{Q} = \mathsf{C}^{-1}, \qquad \mathsf{U} = [\,\mathbf{1},\; v\,], \tag{21}$$

the construction of Eq. (13) with one more column: U is the $n \times 2$ matrix whose columns are the ones vector and the centered redshifts v. The penalty it charges is

$$\delta^{\mathsf{T}}\mathsf{P}_{\text{mem}}\,\delta \;=\; \min_{c,\,s_g} \big(\delta - c\,\mathbf{1} - s_g\,v\big)^{\mathsf{T}}\mathsf{Q}\big(\delta - c\,\mathbf{1} - s_g\,v\big)\,: \tag{22}$$

the prior subtracts the best straight line, weighted by C^{-1}, and charges only the residual. A monotonic drift of any size costs nothing; bumps around the line still cost, and still forget over the length L. Q is tridiagonal (Appendix I), a local object that links neighbors, while the subtracted correction has rank two and is dense, because the fitted line listens to every bin. P_{mem} annihilates both columns, $\mathsf{P}_{\text{mem}}\mathsf{U} = 0$, is positive semidefinite, and has rank $n - 2$, so a group of one or two bins has $\mathsf{P}_{\text{mem}} = 0$.

D The bias continuity equation, step by step

The matter density $\rho_m(\mathbf{x}, t)$ defines the matter overdensity $\delta_m = (\rho_m - \bar{\rho}_m)/\bar{\rho}_m$, with $\bar{\rho}_m(t)$ the mean density; the galaxy number density defines the galaxy-count overdensity δ_{gal} the same way. Conservation of mass and of galaxy number gives each field a continuity equation. In comoving coordinates, keeping only terms linear in the fluctuations (the convective term $\mathbf{v}\cdot\nabla\delta$ is second order),

$$\frac{\partial \delta_m}{\partial t} \;=\; -\,\frac{1}{a}\,\nabla\cdot\mathbf{v}_m, \qquad \frac{\partial \delta_{\text{gal}}}{\partial t} \;=\; -\,\frac{1}{a}\,\nabla\cdot\mathbf{v}_{\text{gal}}, \tag{23}$$

with t cosmic time and $\mathbf{v}_m$ and $\mathbf{v}_{\text{gal}}$ the peculiar velocity fields of the matter and of the galaxies.

The two velocity fields are the same. On large scales the galaxies carry no pressure of their own and fall in the same gravitational potential as the matter, so a conserved sample rides the matter flow: $\mathbf{v}_{\text{gal}} = \mathbf{v}_m$, the no-velocity-bias assumption (Tegmark and Peebles [10] state it as galaxies behaving as test particles). Subtracting the two equations and integrating in time from z_c leaves

$$\frac{\partial}{\partial t}\big(\delta_{\text{gal}} - \delta_m\big) \;=\; 0 \quad\Longrightarrow\quad \delta_{\text{gal}}(z) - \delta_m(z) \;=\; \delta_{\text{gal}}(z_c) - \delta_m(z_c)\,: \tag{24}$$

the excess clumping keeps its initial value forever.

Now substitute the bias. At z_c, $\delta_{\text{gal}}(z_c) = b_c\,\delta_m(z_c)$, so the frozen difference is $(b_c - 1)\,\delta_m(z_c)$, and the matter grows as $\delta_m(z) = \delta_m(z_c)\,D(z)/D(z_c)$. Then

$$\delta_{\text{gal}}(z) \;=\; \delta_m(z) + \big(b_c - 1\big)\,\delta_m(z_c) \;=\; \delta_m(z)\left[\,1 + \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,\right], \tag{25}$$

and dividing by $\delta_m(z)$ gives the bias of Eq. (14). Dividing instead by the growth-tied bias $b_c\,D(z_c)/D(z)$ gives the deviation of Eq. (15). As D grows the frozen term is divided by an ever larger number, so $b \to 1$, from above when $b_c > 1$ and from below when $b_c < 1$: gravity debiases a conserved sample, because the part of $\delta_{\rm gal}$ that traces the matter keeps growing while the frozen excess does not.

Tegmark and Peebles drop the assumption that $\delta_{\rm gal}$ is proportional to δ_m point by point. They describe the pair of fields by the matter fluctuation amplitude, the bias b (the ratio of the galaxy to the matter fluctuation amplitude) and the correlation coefficient ρ_{gm} between the two fields; $\rho_{gm} < 1$ is stochastic bias. The same continuity argument evolves both (their section "After the galaxy formation epoch"): with D normalized to one today and the subscript zero marking today,

$$b_0 = \left[(1-D)^2 + 2D(1-D)\,b\,\rho_{gm} + D^2 b^2\right]^{1/2}, \qquad \rho_{gm,0} = \frac{(1-D) + D\,b\,\rho_{gm}}{b_0}, \tag{26}$$

where b and ρ_{gm} are the values at the epoch whose growth factor is D. Both b_0 and $\rho_{gm,0}$ approach one as clustering grows. For $\rho_{gm} = 1$, Eq. (26) collapses to $b_0 - 1 = (b-1)\,D$, which is Eq. (14) again: the conserved population of this note is the perfectly correlated case of their model.

For a DESI sample the assumption that breaks is conservation: the flux limit and color cuts admit and remove galaxies as redshift grows, mergers turn two members into one, and an ELG belongs to the sample only during its star-forming phase (Appendix B.3). The no-velocity-bias and linear-theory assumptions hold well on the scales this forecast keeps. The script of Appendix L integrates Eq. (23) numerically and recovers Eq. (14) to a relative accuracy of 2×10^{-9}.

E The BGS flux-limit bias model, step by step

This appendix derives the BGS drift used by the baseline prior: a 22.3% span of the deviation, a convex shape, and a 2.4% rms residual around the best straight line. The model lives in the function `bgs_flux_limit_bias` of `plots/make_theory_data.py`, and the script of Appendix L checks its numbers.

E.1 The four steps

Step 1: the flux limit becomes a luminosity threshold. BGS Bright keeps every galaxy brighter than apparent magnitude $r = 19.5$ (measured limit [23]; r is the magnitude band here, not a correlation coefficient). The apparent magnitude r and the absolute magnitude M are related by $r = M + 25 + 5\log_{10}[d_L/(h^{-1}\,\mathrm{Mpc})] + K(z)$, with d_L the luminosity distance, h the Hubble constant in units of 100 km/s/Mpc, and $K(z)$ the K-correction, the magnitude shift of a fixed filter sampling a redshifted spectrum [22]. The model sets $K = 0$, so the faintest absolute magnitude that passes the cut at redshift z is

$$M_{\rm lim}(z) - 5\log_{10} h \;=\; 19.5 - 25 - 5\log_{10}\frac{d_L(z)}{h^{-1}\,\mathrm{Mpc}}, \qquad d_L(z) \;=\; (1+z)\,\chi(z), \tag{27}$$

with $\chi(z)$ the comoving distance of flat ΛCDM with $\Omega_m = 0.3$. Every magnitude here carries the $M - 5\log_{10} h$ convention of the M_* in step 3.

Step 2: each bin is a luminosity-threshold sample. At redshift z the sample holds every galaxy brighter than the threshold, $M < M_{\rm lim}(z)$, and the threshold brightens with distance: the one selection $r < 19.5$ becomes a different threshold sample in every bin.

Step 3: the threshold sets the bias. Zehavi et al. [20] measured the large-scale bias of luminosity-threshold samples in the SDSS main sample at $z < 0.25$ (effective redshift about 0.1) and fit it with

$$b(>L_{\rm th}) \;=\; 1.06 + 0.21\,(L_{\rm th}/L_*)^{1.12}, \qquad L_{\rm th}/L_* \;=\; 10^{-0.4\,(M_{\rm lim}-M_*)}, \tag{28}$$

where $L_{\rm th}$ is the threshold luminosity of the sample, and L_* and $M_* - 5\log_{10} h = -20.44$ are the characteristic luminosity and magnitude of the SDSS r-band luminosity function [21]. The model applies the fit at every redshift of the group, without evolution, and reads the published left side $b\,(\sigma_8/0.8)$ as b, with σ_8 the rms matter fluctuation in spheres of $8\,{\rm Mpc}/h$.

Step 4: the deviation from the growth-tied shape. Because B_g is free, only the shape of $b(z)$ matters, and the model fixes the amplitude B by making the mean deviation over the group's bin centers $\bar z_1, \ldots, \bar z_n$ vanish:

$$\delta(\bar z_i) \;=\; \frac{b(\bar z_i)\,D(\bar z_i)}{B} - 1, \qquad B \;=\; \frac{1}{n}\sum_{j=1}^{n} b(\bar z_j)\,D(\bar z_j). \tag{29}$$

At the six $\Delta z = 0.05$ bin centers of BGS, from $z = 0.125$ to 0.375, the threshold brightens from -18.56 to -21.24 (by 2.68 magnitudes), so $L_{\rm th}/L_*$ grows from 0.18 to 2.10 and the bias from 1.09 to 1.54, while D falls from 0.937 to 0.824. The deviation then runs from -8.0% to $+14.4\%$, the 22.3% span drawn in Figure 9.

E.2 Why the drift is convex

Across the bin centers the bias rises by 41% while D falls by 12%, so the deviation inherits the shape of the bias curve. That curve is flat at low redshift and steep at high redshift because $L_{\rm th}/L_*$ is exponential in the threshold magnitude. Below L_* the term $0.21\,(L_{\rm th}/L_*)^{1.12}$ is small against the constant 1.06 (0.03 at the first bin center), so the bias stays near the faint-end floor of the fit. The threshold crosses L_* near $z = 0.27$, and past the crossing the same term dominates and rises exponentially (0.48 at the last center). A curve that is flat on one side of a crossing and exponential on the other is convex.

A straight line through a convex curve leaves a U-shaped residual. The deviation at the six centers is almost exactly a parabola: adding a quadratic term cuts the residual from 2.4% to 0.1% rms. The fitted quadratic coefficient is 3.8 per unit redshift squared, and a pure parabola with that coefficient, sampled at the six centers with its best line removed, leaves $+3.2\%$ at the ends and -2.6% in the middle, rms 2.4%: the measured residuals $(+3.1, -0.5, -2.5, -2.6, -0.8, +3.3)\%$ are this U shape. The 2.4% rms is therefore the curvature of the drift, the one part of it a free tilt cannot absorb.

E.3 What the model leaves out

None of the omissions below changes the qualitative shape, a smooth, monotonic, convex rise of the bias toward the far edge of the group; each states the direction in which it pushes the drift.

The threshold-bias relation is frozen at $z \approx 0.1$. Two neglected evolutions pull in opposite directions. The bias at fixed luminosity grows with redshift ($b - 1 \propto 1/D$ for a conserved population, Eq. (14)), which steepens the drift. Galaxies were also brighter in the past: Blanton et al. [21] measure a brightening of M_* by 1–2 magnitudes per unit redshift, so a fixed threshold sits lower relative to $L_*(z)$ at high redshift, which flattens the drift. The flattening is a fraction of the effect: the brightening reaches 0.3–0.6 magnitudes at the last bin center, against the 2.68 magnitudes of the threshold itself.

No K-correction. The bandpass-shift part alone, for a spectrum flat in wavelength, is $2.5 \log_{10}(1 + z)$, or 0.37 magnitudes at $z = 0.4$ [22], and a red spectrum makes the r-band correction larger still. The M_* of step 3 is also measured in the r band shifted to $z = 0.1$ [21], while Eq. (27) produces an observed-band magnitude; the K-correction is the bridge the model omits. A positive K makes the true threshold brighter at high redshift, so the omission steepens the drift.

Only $r < 19.5$ is modeled. BGS Bright also imposes a fiber-magnitude cut, and BGS Faint, the color-selected extension at $19.5 < r < 20.175$, is absent from the model [23]. Including BGS Faint lowers the limit by 0.675 magnitudes at every redshift, which lowers $L_{\rm th}/L_*$ and the bias everywhere and moves the L_* crossing to higher redshift: a flatter drift. The fiber cut and the color selection change the galaxy mix in a way Eq. (28) cannot follow, and the model assigns them no direction.

Scope. The model is an order-of-magnitude estimate, not a prediction of the measured DESI BGS bias. Its reliable content is the shape of the drift (smooth, monotonic, convex) and its size to a factor of order unity, and that is the content this note uses: the baseline prior takes $\sigma_g = 3\%$ for BGS from the 2–3% curvature.

F The accuracy test

This appendix gives the method of Section 5: the shift of a fit under a known drift, the two drift laws, the random truths built on them, and the statistic $\sqrt{\langle \chi^2 \rangle / n_{\rm dof}}$ of Eq. (6), whose value is written η.

F.1 The shift formula

Let m be the expected data vector (the band powers), C_d its covariance, D the matrix of derivatives of m with respect to the parameters (sans-serif, not the growth factor D), and $F = \mathsf{D}^\mathsf{T} \mathsf{C}_d^{-1} \mathsf{D} + F_{\rm priors}$ the Fisher matrix of one fit, its priors included (Appendix K); under model A the parameters include the variant's higher-order parameters, marginalized like the cosmology. If the true biases carry a drift the fit may not express, the data vector moves by δm, and to first order the maximum of the posterior moves by

$$\delta p \;=\; F^{-1} \mathsf{D}^\mathsf{T} \mathsf{C}_d^{-1}\, \delta m. \tag{30}$$

The derivation is one line: the score (the gradient of the log posterior) at the old maximum becomes $\mathsf{D}^\mathsf{T} \mathsf{C}_d^{-1} \delta m$ instead of zero, and the curvature that converts a score into a displacement

is F. With C_w the fit's marginalized 2×2 covariance of (w_0, w_a) and δw the (w_0, w_a) entries of δp, the shift is

$$\Delta_w \;=\; \sqrt{\delta w^{\mathsf{T}}\, \mathsf{C}_w^{-1}\, \delta w}, \tag{31}$$

the displacement of the dark-energy ellipse in units of its own size; Ω_m and A_s shifts are $|\delta p|/\sigma_p$. The threshold drawn in the figures is $1/3$, a shift that widens the effective error by a few percent. Both the ellipse and the response use the same fit's F, so a rigid model is punished twice: it absorbs little of the drift, and its ellipse is small.

The drift enters through the bin biases: with β_i the amplitudes of Eq. (9) and δ_i the drift's fractional deviations, $\delta m = \sum_i (\partial m/\partial \beta_i)\, B_g\, \delta_i$. Under model A the column $\partial m/\partial \beta_i$ is the chain-rule column of Eq. (64) divided by $D(\bar{z}_i)$, so a drift of b_1 drags b_2 and b_{s^2} along through their calibrations. For the growth-tied prior the response is evaluated in its own parameter basis; for the free prior the bias derivatives span the drift exactly, so the cosmology shift vanishes, and the accuracy of that prior is bought entirely with the information it gives up.

F.2 The drift laws

Two known drifts are injected. The conserved-population drift applies to every group: the deviation of Eq. (15), anchored at the group's mean redshift, nearly a straight line of a few percent across each group. The BGS flux-limit drift applies to BGS alone: the convex deviation of Appendix E, a 22% span whose best line leaves a 2.4% rms curvature. Both come from `plots/make_theory_data.py`; only their shapes matter, because the free amplitudes remove the level.

F.3 Random truths

A random truth sets the bias of bin i in group g to

$$b_i \;=\; \frac{B_g}{D(\bar{z}_i)} \left(1 + \delta_i^{\mathrm{drift}} + \varepsilon_i\right) \tag{32}$$

at the fiducial cosmology. The drift $\delta_i^{\mathrm{drift}}$ is the known part: the flux limit in BGS and the conserved population in the other five groups. The residual ε_i is one draw from the Gaussian $\mathcal{N}(0, \Sigma)$ whose covariance is the kernel of one of the two stated priors, $\Sigma_{ij} = \sigma_g^2\, e^{-|\bar{z}_i - \bar{z}_j|/L}$ inside a group and zero across groups. Two families of 2000 truths are drawn, baseline and pessimistic scatter, and every truth is fitted under every prior, each centered on the growth-tied shape $\delta = 0$. No fit is told which truth it faces: the baseline prior meets truths drawn from its own prior and truths drawn from the pessimistic prior, and is judged on both.

Because Eq. (30) is linear in δm, and δm is linear in the bias deviations, the (w_0, w_a) rows of the response collapse into one $2 \times n_{\mathrm{bins}}$ matrix A per prior, layout and variant, with n_{bins} the number of bins of the layout:

$$\delta w \;=\; \mathsf{A}\, \delta, \qquad \delta_i \;=\; \delta_i^{\mathrm{drift}} + \varepsilon_i, \tag{33}$$

so the 2000 truths cost one matrix product. The draws check themselves against the closed form $\langle \Delta_w^2 \rangle = \Delta_{\mathrm{drift}}^2 + \mathrm{tr}(\mathsf{C}_w^{-1} \mathsf{A} \Sigma \mathsf{A}^{\mathsf{T}})$, with Δ_{drift} the shift of the drift alone. The response is first order in δ, while the band powers are quadratic in the bin biases: the dropped term is a few percent of the shift for the 1–3% residuals and perhaps 10–20% for the 22% BGS drift.

Where the fit's prior equals the truth's scatter (the matched case), the average shift is not a bias. That fit's ellipse is wider than the growth-tied one precisely because it marginalizes over the scatter the truth now has: the fitted (w_0, w_a) moves from draw to draw, and the ellipse was built to cover those moves. Whether an ellipse is honest is decided by the statistic of the next subsection.

F.4 The test statistic

A fit is correctly specified when its quoted ellipse covers the truth with the stated probability. The squared distance of the fitted (w_0, w_a), written $\hat{w}$, from the truth $w_{\rm true}$ in units of the ellipse is $\chi^2_w = (\hat{w} - w_{\rm true})^{\mathsf{T}} \mathsf{C}_w^{-1} (\hat{w} - w_{\rm true})$. Averaged over the bias history and the data noise,

$$\langle \chi^2_w \rangle \;=\; \Delta^2_{\rm drift} \;+\; \mathrm{tr}\Big(\mathsf{C}_w^{-1} \mathsf{A}\, \Sigma\, \mathsf{A}^{\mathsf{T}}\Big) \;+\; \mathrm{tr}\big(\mathsf{C}_w^{-1} \mathsf{C}_w^{\rm noise}\big)\,: \tag{34}$$

the drift squared, the scatter that the fit's bias parameters, held back by their prior, did not follow, and the noise. Here $\mathsf{C}_w^{\rm noise}$ is the share of C_w contributed by the data noise and by the Gaussian priors other than the bias prior (the velocity dispersions and the higher-order parameters); the cross terms average to zero. Eq. (34) is evaluated in closed form: it is the exact average over infinitely many universes, each with its own noise realization and bias history. The 2000 draws only make the percentile bars of Figure 2, and they carry no noise. The quoted value is $\eta = \sqrt{\langle \chi^2_w \rangle / 2}$.

An honest ellipse gives $\langle \chi^2_w \rangle = 2$: if the fitted (w_0, w_a) scatters around the truth with covariance exactly C_w, the expected squared distance is $\mathrm{tr}(\mathsf{C}_w^{-1} \mathsf{C}_w) = 2$, one unit per parameter. Growing both axes of the ellipse by η divides χ^2_w by η^2 and brings the average back to 2; this matches the coverage exactly when the real scatter of $\hat{w}$ has the shape and orientation of the ellipse.

Coverage of one parameter. For one parameter p, in units of its marginalized error σ_p, the same three terms read

$$\langle \chi^2_p \rangle \;=\; \mu^2 \;+\; \mathsf{A}_p^{\mathsf{T}} \Sigma\, \mathsf{A}_p \;+\; \nu, \tag{35}$$

with μ the shift of the known drift alone, A_p the row of the response that carries the bias deviations into p, divided by σ_p, and ν the noise term: the (p,p) entry of $F^{-1} F_0 F^{-1}$ over σ_p^2, where $F_0 = \mathsf{D}^{\mathsf{T}} \mathsf{C}_d^{-1} \mathsf{D} + F_{\rm other}$ holds the data and the Gaussian priors other than the bias prior. The bias prior adds a positive semidefinite precision, $F \succeq F_0$, so $F^{-1} F_0 F^{-1} \preceq F^{-1} F F^{-1} = F^{-1}$, whose (p,p) entry is σ_p^2: $\nu \le 1$ (with $X \preceq Y$ meaning $x^{\mathsf{T}} X x \le x^{\mathsf{T}} Y x$ for every x). The rms displacement beyond the noise, $(\mu^2 + \mathsf{A}_p^{\mathsf{T}} \Sigma \mathsf{A}_p)^{1/2}$, is therefore at least $(\eta^2 - 1)^{1/2}$, with $\eta = \langle \chi^2_p \rangle^{1/2}$.

Because the response is linear and both the noise and the residual are Gaussian, the pull $\zeta = (\hat{p} - p_{\rm true})/\sigma_p$ is exactly Gaussian, with mean μ and variance $V = \langle \chi^2_p \rangle - \mu^2$. The 1σ bar covers the truth when $|\zeta| < 1$, in a fraction

$$P(|\zeta| < 1) \;=\; \Phi\!\left(\frac{1-\mu}{\sqrt{V}}\right) - \Phi\!\left(\frac{-1-\mu}{\sqrt{V}}\right) \tag{36}$$

of the universes, with $\Phi(x) = \frac{1}{2}\big[1 + \mathrm{erf}(x/\sqrt{2})\big]$ the cumulative distribution of a unit Gaussian. When the prior absorbs the known drift ($\mu \approx 0$) the coverage depends on η alone,

$$P(|\zeta| < 1) \;=\; 2\,\Phi(1/\eta) - 1\,: \tag{37}$$

68% at $\eta = 1$, 56% at 1.3, 50% at 1.5, 38% at 2 and 31% at 2.5. For $\eta \geq 1$, $\mu = 0$ is the most generous split of $\langle \chi_p^2 \rangle$ between drift and variance, so otherwise Eq. (37) is an upper bound. For the baseline, pessimistic and free priors ($\mu \leq 0.33$) it is exact to 0.1%; for the growth-tied prior only Eq. (36) is right.

Under the pessimistic scatter and model A baseline, the bar covers the truth, on $\Delta d_{\rm com} =$ 62 Mpc, $\Delta z = 0.02$, 0.05 and 0.2, in 56, 63, 40 and 68% of universes for Ω_m and 57, 65, 54 and 68% for A_s under the baseline prior; 66–68% under the pessimistic prior; and 68.3% under the free prior. Under the growth-tied prior it is 21, 39, 5 and 7% for Ω_m, where Eq. (37) would give 24, 39, 19 and 22%. For (w_0, w_a) the ellipse $\chi_w^2 < 2.30$ contains 68.3% of a two-dimensional Gaussian, but the exact coverage depends on how the excess of $\langle \chi_w^2 \rangle$ splits between the two directions, so only η is quoted.

The baseline prior against a pessimistic truth. The baseline prior states that each group's residual around its free line is small and smooth ($\sigma_g = 1\%$, 3% for BGS, coherent over $\Delta z \approx 0.2$); the pessimistic truth has 3% wiggles that change over $\Delta z \approx 0.05$. The prior charges a large penalty for bin biases that follow such wiggles, so the biases follow them only in part, and the fit moves the cosmological parameters to absorb the rest of each wiggle. The push changes from truth to truth, so over the 2000 truths Δ_w spreads over the percentile bars of Figure 2, and the observed universe is one of these displacements, reported with the baseline-prior ellipse. At $\eta = 2$, for example, the real scatter of the best-fit (w_0, w_a) is twice as wide per axis as the reported ellipse. The FoM, proportional to the inverse area, would then be four times too large; the rescaled FoM is still not a number to quote, because the ellipse's center and shape come from a prior narrower and smoother than the truth. If the pessimistic truth is possible, the fit to report is the pessimistic prior. The opposite mismatch, a baseline truth fitted with the pessimistic prior, gives η below 1: a conservative ellipse.

The matched case. The matched case lands on 1 by an identity. Collect all parameters in θ, with every prior centered on the fiducial $\theta_{\rm fid}$, and let $\Delta\theta$ be the truth's departure from it: zero in the cosmology and in every parameter with another Gaussian prior, $B_g\,\delta_i$ in the amplitude β_i. With $\mathbf{n}_d$ the data noise, of covariance C_d, the linear response puts the fit at $\hat\theta = \theta_{\rm fid} + F^{-1}\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}(\mathsf{D}\,\Delta\theta + \mathbf{n}_d)$. Write $F = \mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\mathsf{D} + F_{\rm bias} + F_{\rm other}$, with $F_{\rm bias}$ the bias prior's precision (the projection of Eq. (21) in the amplitude block). Since $F_{\rm other}\,\Delta\theta = 0$, the fit's error is

$$\hat\theta - \theta_{\rm true} \;=\; F^{-1}\big(\mathsf{D}^\mathsf{T}\mathsf{C}_d^{-1}\mathbf{n}_d \;-\; F_{\rm bias}\,\Delta\theta\big)\,: \tag{38}$$

the noise, plus the part of the truth's departure that the bias prior held back. Treating each other Gaussian prior as one more independent measurement, the noise part has covariance $F^{-1}F_0F^{-1}$. In the matched case $\mathsf{P}_{\rm mem}$ is built from the same kernel Σ as the truth's scatter (in the units of the bin amplitudes, a factor B_g^2 per group), and every projection of that form obeys

$$\mathsf{P}_{\rm mem}\,\Sigma\,\mathsf{P}_{\rm mem} \;=\; \mathsf{P}_{\rm mem}. \tag{39}$$

The proof takes one line: with $\mathsf{Q} = \Sigma^{-1}$ and $\mathsf{M} = (\mathsf{U}^\mathsf{T}\mathsf{Q}\mathsf{U})^{-1}$, $\Sigma\,\mathsf{P}_{\rm mem} = \mathsf{I} - \mathsf{U}\mathsf{M}\mathsf{U}^\mathsf{T}\mathsf{Q}$, and $\mathsf{P}_{\rm mem}\mathsf{U} = 0$ removes the second term. The scatter part of Eq. (38) then has covariance $F^{-1}F_{\rm bias}F^{-1}$, the two covariances add up to $F^{-1}FF^{-1} = F^{-1}$, and the scatter and noise terms of Eq. (34) sum to exactly 2. The widening the prior adds to the ellipse equals the error the scatter inflicts. The matched value is $\eta = (1 + \Delta^2_{\rm drift}/2)^{1/2}$ for (w_0, w_a) and $(1 + \Delta^2_{{\rm drift},p})^{1/2}$

for one parameter, so it departs from 1 only through the known drift: under model A it stays within 3% of 1 for (w_0, w_a), while a single parameter reaches 1.05 (Ω_m on $\Delta z = 0.05$, where the BGS flux limit is absorbed least well). Appendix O verifies the identity numerically.

When a fit is conservative. Write $\mathsf{P}_{\rm mem}$ in its null-space form, Eq. (59), with the fit's kernel $\Sigma_{\rm fit}$, and let $\Sigma_{\rm true}$ be the truth's. If the truth's scatter is no larger than the fit's prior in every combination the prior charges, $\mathsf{N}^{\mathsf{T}}\Sigma_{\rm true}\mathsf{N} \preceq \mathsf{N}^{\mathsf{T}}\Sigma_{\rm fit}\mathsf{N}$, then $\mathsf{P}_{\rm mem}\Sigma_{\rm true}\mathsf{P}_{\rm mem} \preceq \mathsf{P}_{\rm mem}$, the error covariance is at most F^{-1}, and $\eta \le (1+\Delta^2_{\rm drift}/2)^{1/2}$: a prior wider than the truth wastes information but does not understate the error. Scatter larger or rougher than the pessimistic prior would make the pessimistic prior's error bar too small in the same way as above; the ladder ends at the free prior, for which $F_{\rm bias} = 0$ leaves only the noise in Eq. (38), so $\eta = 1$ for every truth expressible through the bin biases.

F.5 The same test for Ω_m and A_s

The same drifts and residuals are scored for Ω_m and A_s, each in units of its own marginalized error, with $\chi^2_p = (\hat{p} - p_{\rm true})^2/\sigma_p^2$ and $\eta = \langle\chi^2_p\rangle^{1/2}$, the case $n_{\rm dof} = 1$ of Eq. (6); Eq. (36) converts it into coverage. These two can react more strongly to a bias error than (w_0, w_a), because a bias error moves the clustering amplitude $b_i^2 A_s D^2(\bar{z}_i)$ of bin i, which A_s scales directly and Ω_m moves through the growth factor.

G The projected precision for two bins

Everything essential already happens with two bins. Call their amplitudes β_1 and β_2, with common fiducial value B, prior spread σ_β per bin, and correlation coefficient r between the two deviations. The prior covariance and its inverse are

$$\mathsf{C} \;=\; \sigma_\beta^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix} \tag{40}$$

(the off-diagonal of $\mathsf{C}\,\mathsf{C}^{-1}$ is $r - r = 0$ and the diagonal $(1-r^2)/(1-r^2) = 1$). The prior penalty of a displacement $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^{\mathsf{T}}$ is $\Delta\beta^{\mathsf{T}}\,\mathsf{C}^{-1}\Delta\beta$, and two kinds of move matter.

A uniform shift. For $\Delta\beta = c\,(1,1)^{\mathsf{T}}$, both bins up by c,

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{c^2\left[(1-r)+(1-r)\right]}{\sigma_\beta^2(1-r^2)} \;=\; \frac{2c^2}{\sigma_\beta^2\,(1+r)} \;\neq\; 0. \tag{41}$$

The plain prior charges for a uniform shift, but a uniform shift is a change of B_g, which the growth-tied model leaves free; keeping the penalty would constrain B_g and make the comparison unfair.

A shape change. For $\Delta\beta = d\,(1,-1)^{\mathsf{T}}$, the bins moving apart,

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{2d^2}{\sigma_\beta^2\,(1-r)}. \tag{42}$$

This is the penalty the prior is meant to apply: it grows without bound as $r \to 1$, where the two bins are forced to track each other, and relaxes to $2d^2/\sigma_\beta^2$ as $r \to 0$. At the worked value $r = 0.670$ of Appendix A.2, a shape change costs $6.1\, d^2/\sigma_\beta^2$ and a uniform shift of the same size $1.2\, c^2/\sigma_\beta^2$: the correlated prior already charges five times more for the shape, and the projection takes the uniform charge to zero.

The projected precision. Both rows of C^{-1} have the same sum, so $\mathsf{C}^{-1}\mathbf{1} = [\sigma_\beta^2(1+r)]^{-1}(1,1)^\mathsf{T}$ and $\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1} = 2/[\sigma_\beta^2(1+r)]$. Substituting into Eq. (13),

$$\mathsf{P} \;=\; \frac{1}{2\sigma_\beta^2(1-r)}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^\mathsf{T}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1-\beta_2)^2}{2\sigma_\beta^2(1-r)}. \tag{43}$$

The projected prior charges only the difference of the two amplitudes: a uniform shift is free, and a shape change keeps the penalty of Eq. (42). The denominator is the prior variance of the jump, $\mathrm{Var}(\beta_1-\beta_2) = 2\sigma_\beta^2(1-r)$, so the construction reduces to one sentence: *the prior constrains the jump, not the level.*

H The projection as a marginalization

Equation (13) subtracts from C^{-1} a matrix built from the single vector $\mathsf{C}^{-1}\mathbf{1}$, so it removes the prior's information about one direction, the uniform shift, and leaves everything else alone. Equivalently, the projected penalty of a displacement $\Delta\beta$ is the plain penalty of what remains after subtracting the best uniform shift, the generalized-least-squares mean $\hat{c} = \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\Delta\beta/\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}$. The derivation below shows why this is the correct prior and not just one possible repair. A group has n bins throughout.

H.1 Marginalizing a flat-prior amplitude

Model the group amplitudes as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{44}$$

with a flat prior on the scalar B: the model of Eq. (7), linearized around the fiducial. The prior density of β follows by integrating B out,

$$p(\beta) \;\propto\; \int dB\, \exp\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^\mathsf{T}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{45}$$

The exponent is quadratic in B. With $S = \mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}$, completing the square gives

$$\big(\beta - B\mathbf{1}\big)^\mathsf{T}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^\mathsf{T}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\beta}{S}. \tag{46}$$

The B integral is a Gaussian centered on B_* with width $1/\sqrt{S}$, and contributes only the β-independent constant $\sqrt{2\pi/S}$. What survives is

$$p(\beta) \;\propto\; \exp\Big[-\tfrac{1}{2}\,\beta^\mathsf{T}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}}{\mathbf{1}^\mathsf{T}\mathsf{C}^{-1}\mathbf{1}}, \tag{47}$$

exactly the projection of Eq. (13); B_* is the same generalized-least-squares mean as $\hat{c}$, because the best-fitting amplitude dominates the integral.

The same computation has a standard matrix name. The exponent of Eq. (45) is a Gaussian in the joint variables (β, B) with the $(n+1)\times(n+1)$ precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1} & \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}. \tag{48}$$

Removing a parameter from a precision matrix is not just deleting its row and column: the remaining block must be corrected downward by whatever the removed parameter could absorb. The corrected block, $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$, is the *Schur complement* over B, and it reproduces Eq. (13) term by term. So P is precisely the statement "*B_g is sampled with a flat prior and marginalized*", folded into one matrix.

I Many bins: the tridiagonal inverse

I.1 Three bins: total correlation versus direct links

A group has n bins at redshifts $\bar{z}_1 < \cdots < \bar{z}_n$, unevenly spaced in general, with the prior covariance $\mathsf{C}_{ij} = \sigma_\beta^2 \exp(-|\bar{z}_i - \bar{z}_j|/L)$ of Eq. (10). This matrix is positive definite for distinct redshifts, so C^{-1} exists. Its structure shows on three evenly spaced bins: with neighbor correlation $r = e^{-\Delta z/L}$ the outer pair carries r^2, and

$$\mathsf{C} \;=\; \sigma_\beta^2 \begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)} \begin{pmatrix} 1 & -r & 0 \\ -r & 1+r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{49}$$

Row 1 of C^{-1} times column 3 of C gives, without the prefactors, $1\cdot r^2 - r\cdot r + 0\cdot 1 = 0$, as the off-diagonal of the identity must. Bins 1 and 3 *are* correlated (C contains r^2), yet their entry in C^{-1} is zero: C gives the total correlation, what δ_1 says about δ_3 when nothing else is known, while C^{-1} gives the direct one, what δ_1 still says once δ_2 is known, and that is nothing. The exponential kernel makes this exact because its correlations multiply along the chain,

$$e^{-(\bar{z}_3-\bar{z}_1)/L} \;=\; e^{-(\bar{z}_3-\bar{z}_2)/L}\; e^{-(\bar{z}_2-\bar{z}_1)/L}, \tag{50}$$

so the middle bin already carries everything the outer bins share. A Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize this way, because squared separations do not add along the chain, and its inverse is dense.

I.2 The chain of links

Because correlations multiply along the chain, the prior can be built one bin at a time: the joint density of the deviations factorizes into neighbor conditionals,

$$\begin{aligned} p(\delta_1,\ldots,\delta_n) \;&=\; p(\delta_1)\,p(\delta_2 \mid \delta_1)\cdots p(\delta_n \mid \delta_{n-1}), \\ \big\langle \delta_{k+1} \mid \delta_k \big\rangle = r_k\,\delta_k, \qquad & \mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) = \sigma_g^2\,\big(1-r_k^2\big), \end{aligned} \tag{51}$$

with $r_k = e^{-(\bar{z}_{k+1}-\bar{z}_k)/L}$ the correlation of the neighbors k and $k+1$. The conditional follows from the covariances alone. Split $\delta_{k+1} = \lambda\,\delta_k + e$ and choose λ so that the leftover e carries

no correlation with δ_k: $\mathrm{Cov}(e,\delta_k) = \sigma_g^2 r_k - \lambda\,\sigma_g^2 = 0$ gives $\lambda = r_k$, and then $\mathrm{Var}(e) = \sigma_g^2 - 2r_k^2\sigma_g^2 + r_k^2\sigma_g^2 = \sigma_g^2(1-r_k^2)$. For jointly Gaussian variables uncorrelated means independent, so knowing the neighbor explains the fraction r_k^2 of the bin's variance and the fraction $1-r_k^2$ is new. Earlier bins add nothing: for $j \le k$,

$$\mathrm{Cov}(e,\delta_j) \;=\; \sigma_g^2\, e^{-(\bar z_k - \bar z_j)/L}\left[e^{-(\bar z_{k+1}-\bar z_k)/L} - r_k\right] \;=\; 0, \tag{52}$$

because the correlation factorizes through $\bar z_k$. Taking $-2\ln p$ gives the penalty as a sum over neighbor links,

$$\delta^{\mathsf T}\big(\sigma_g^2\mathsf R\big)^{-1}\delta \;=\; \frac{\delta_1^2}{\sigma_g^2} \;+\; \sum_{k=1}^{n-1}\frac{\big(\delta_{k+1} - r_k\,\delta_k\big)^2}{\sigma_g^2\,\big(1-r_k^2\big)}\,: \tag{53}$$

each term compares one bin with r_k times the bin before it. *Directly*, only neighbors talk; *in total*, the influence propagates down the whole chain. Probability theory calls this the Markov property.

I.3 The tridiagonal inverse

Expanding each square of Eq. (53) and collecting the coefficient of every product $\delta_i\delta_j$ reads off the precision matrix. δ_1^2 appears in the first term and in the $k=1$ link as the predictor, $1 + r_1^2/(1-r_1^2) = 1/(1-r_1^2)$; an interior δ_k^2 appears in link $k-1$ as the predicted bin and in link k as the predictor; the last bin is only ever predicted. Each product $\delta_k\delta_{k+1}$ appears in its own link only, and no product of non-neighbors appears. For the amplitudes ($\Delta\beta = B_g\,\delta$, $\sigma_\beta = \sigma_g B_g$) the inverse is therefore tridiagonal,

$$\sigma_\beta^2\,\big(\mathsf C^{-1}\big)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1,\\[2mm] \dfrac{1}{1-r_{k-1}^2} + \dfrac{r_k^2}{1-r_k^2}, & 1<k<n,\\[2mm] \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad \sigma_\beta^2\,\big(\mathsf C^{-1}\big)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}. \tag{54}$$

The off-diagonal entry is the link of pair k, stronger when the bins are closer; the diagonal entry of a bin collects the links touching it. For even spacing, all $r_k = r$, the interior diagonal becomes $(1+r^2)/(1-r^2)$ and

$$\mathsf C^{-1} \;=\; \frac{1}{\sigma_\beta^2\,(1-r^2)}\begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{55}$$

which for $n=3$ is Eq. (49).

I.4 The projection, group by group, and its limits

Each group receives the projected precision of Eq. (13), built from its own $\mathsf C$; across groups the prior is zero, so the total prior precision is block diagonal. $\mathsf P\,\mathbf 1 = 0$, $\mathsf P$ is positive semidefinite

(no displacement is rewarded), and it has rank $n-1$: of the n directions a group can move in, exactly the uniform shift is free.

The kernel model contains both extremes. As $L \to \infty$, every $r_k \to 1$, the jump variance $2\sigma_g^2(1-r_k)$ of every neighbor pair vanishes, and only the uniform shift is left: the growth-tied model, whatever σ_g. As $\sigma_g \to \infty$ at fixed L, $\mathsf{C}^{-1} \to 0$ and $\mathsf{P} \to 0$: the free model.

J The memory projection, step by step

Throughout, a group has n bins, C is the kernel covariance $\mathsf{C}_{ij} = \sigma_g^2 e^{-|\bar{z}_i - \bar{z}_j|/L}$ of the residuals, $\mathsf{Q} = \mathsf{C}^{-1}$, $\mathsf{U} = [\,\mathbf{1},\ v\,]$ is the $n \times 2$ design matrix of Eq. (21) with v the centered redshifts, and $\theta = (c, s_g)^{\mathsf{T}}$ collects a line's level and slope. (For the bin amplitudes every covariance carries a factor B_g^2 that cancels in each identity below.)

J.1 The line fit, the split, and the marginalization

Minimizing the penalty of a displacement δ over the line, $\chi^2(\theta) = (\delta - \mathsf{U}\theta)^{\mathsf{T}}\mathsf{Q}\,(\delta - \mathsf{U}\theta)$, gives the normal equations of a generalized least-squares fit,

$$\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\,\hat{\theta} \;=\; \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta, \qquad \hat{\theta} = (\hat{c}, \hat{s}_g)^{\mathsf{T}}, \tag{56}$$

with $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U}$ positive definite whenever the group has two distinct redshifts. Writing $\delta = \mathsf{U}\hat{\theta} + \delta_\perp$, the normal equations say $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\delta_\perp = 0$: the residual is orthogonal to both columns in the Q metric. The cross terms vanish and the penalty splits,

$$\delta^{\mathsf{T}}\mathsf{Q}\,\delta \;=\; \hat{\theta}^{\mathsf{T}}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)\hat{\theta} \;+\; \delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp, \qquad \delta^{\mathsf{T}}\mathsf{P}_{\mathrm{mem}}\,\delta \;=\; \delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp : \tag{57}$$

the projection keeps the residual's penalty and discards the line's. A pure line has $\delta_\perp = 0$, so $\mathsf{P}_{\mathrm{mem}}\mathsf{U} = 0$; the residuals obey two linear constraints, so $\mathsf{P}_{\mathrm{mem}}$ has rank $n-2$; and $\delta_\perp^{\mathsf{T}}\mathsf{Q}\,\delta_\perp \geq 0$ makes it positive semidefinite.

Marginalization gives the same matrix, as for the single column $\mathbf{1}$ of Appendix H. With $\delta \sim \mathcal{N}(\mathsf{U}\theta, \mathsf{C})$ and $\theta \sim \mathcal{N}(0, \Lambda)$, $\Lambda = \mathrm{diag}(W^2, \tau^2)$ the prior variances of the level and the slope, the marginal covariance of δ is $\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}$, and the Woodbury identity inverts it:

$$\big(\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}\big)^{-1} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\Lambda^{-1} + \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}. \tag{58}$$

As the prior widths grow, $\Lambda^{-1} \to 0$ and the right-hand side becomes $\mathsf{P}_{\mathrm{mem}}$; the leading correction carries Λ^{-1}, so the deviation falls as $1/W^2$.

J.2 The null-space form

Let N be an $n \times (n-2)$ matrix whose columns span the combinations of bins that no straight line can produce: $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$, plain Euclidean orthogonality to $\mathbf{1}$ and to v. Then

$$\mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q} \;=\; \mathsf{N}\big(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N}\big)^{-1}\mathsf{N}^{\mathsf{T}}. \tag{59}$$

Both sides annihilate U: the left by construction, the right because $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$. Both sides map $\mathsf{C}\mathsf{N}$ to N: the right because $\mathsf{N}(\mathsf{N}^{\mathsf{T}}\mathsf{C}\mathsf{N})^{-1}(\mathsf{N}^{\mathsf{T}}\mathsf{C}\mathsf{N}) = \mathsf{N}$, the left because $\mathsf{Q}\mathsf{C}\mathsf{N} = \mathsf{N}$ and the subtracted term meets $\mathsf{U}^{\mathsf{T}}\mathsf{N} = 0$. And the columns of $\mathsf{C}\mathsf{N}$ and of U together span the whole

space: if $\mathsf{C}\mathsf{N}x + \mathsf{U}y = 0$, multiplying by N^{T} gives $\mathsf{N}^{\mathsf{T}}\mathsf{C}\mathsf{N}\,x = 0$, so $x = 0$, and then $y = 0$. Two matrices that agree on a spanning set are equal. The prior therefore reads off the combinations $\mathsf{N}^{\mathsf{T}}\delta$ that no line can make and charges each by the inverse of its own prior covariance. With the single column $\mathsf{U} = \mathbf{1}$ the same identity gives the kernel projection of Eq. (13), with N spanning the differences between bins.

J.3 Worked examples

One or two bins are fully free. For $n = 2$ the matrix U is square and invertible, so the subtracted term collapses to Q and the projection vanishes; for $n = 1$, N has no columns. Three evenly spaced bins show what survives. With spacing Δ and $r = e^{-\Delta/L}$, $v = (-\Delta, 0, \Delta)^{\mathsf{T}}$, and the only direction orthogonal to $\mathbf{1}$ and v is the discrete curvature $\mathsf{N} = (1, -2, 1)^{\mathsf{T}}$, whose prior variance is $\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N} = \sigma_g^2(6 - 8r + 2r^2) = 2\sigma_g^2(1 - r)(3 - r)$ (the diagonal gives $1 + 4 + 1$, the two neighbor pairs $-8r$, the outer pair $+2r^2$). The penalty is

$$\delta^{\mathsf{T}}\mathsf{P}_{\mathrm{mem}}\,\delta \;=\; \frac{\big(\delta_1 - 2\delta_2 + \delta_3\big)^2}{2\,\sigma_g^2\,(1 - r)(3 - r)}\,: \tag{60}$$

the memory model charges only the curvature and is blind to any straight-line drift, where the kernel model of Eq. (49) charges the two neighbor differences and so taxes a drift as heavily as a zigzag of the same step.

J.4 The universal-kriging predictor

Predict bin $k + 1$ from the first $k \geq 2$ bins, with flat priors on θ. Write C_k for the past kernel block, U_k for the past rows of U, $u_{k+1} = (1, v_{k+1})^{\mathsf{T}}$ for the next row, and γ for the vector of kernel covariances between bin $k + 1$ and the past. Given θ, the Gaussian conditional is

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_{1..k}, \theta\big] \;=\; u_{k+1}^{\mathsf{T}}\theta \;+\; \gamma^{\mathsf{T}}\mathsf{C}_k^{-1}\big(\delta_{1..k} - \mathsf{U}_k\,\theta\big). \tag{61}$$

With a flat prior the posterior of θ is centered on the fit $\hat{\theta}$ of Eq. (56) applied to the past block, and Eq. (61) is linear in θ, so averaging replaces θ by $\hat{\theta}$. For the exponential kernel the weights collapse: by Eq. (50) every entry of γ is r_k times the corresponding entry of C_k's last column, so $\gamma^{\mathsf{T}}\mathsf{C}_k^{-1} = r_k\,e_k^{\mathsf{T}}$, with e_k the k-th unit vector. Substituting gives Eq. (20): the fitted line at the new bin plus one Markov step from the last residual.

J.5 The finite-slope correlation

With a Gaussian slope of variance τ^2 and the level held in B_g, the deviations have the covariance of Eq. (18), so $\mathrm{Var}(\delta_i) = \tau^2 v_i^2 + \sigma_g^2$. For separations much larger than L the kernel term is exponentially small, and dropping it gives the correlation of Eq. (19). When $\tau|v_i|$ and $\tau|v_j|$ both exceed σ_g, the σ_g^2 terms become negligible and the coefficient tends to $v_i v_j/|v_i v_j| = \pm 1$.

K How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix,

$$F \;=\; F_{\mathrm{data}} \;+\; F_{\mathrm{priors}}, \tag{62}$$

with $F_{\rm priors}$ holding one block per prior. The projected precision of each group, $\mathsf{P}_{\rm mem}$ of Eq. (21) for the baseline and pessimistic priors, is added to the rows and columns of that group's amplitudes β_i; all other parameters are unchanged. The growth-tied prior is not a block of this kind: it sums each group's amplitude columns into one (Appendix M.4).

Bins that lost all their band powers to the scale cut have zero rows in $F_{\rm data}$. In the free model such a parameter is unconstrained and must be dropped before inversion. With the correlated prior it stays: the prior ties the empty bin to its neighbors, so the matrix stays invertible as long as the group's data fix its free level and tilt. Appendix O describes the pseudo-inverse that guards the one system where a direction carries no information.

L Numerical verification

The companion script `verify_prior.py`, in this folder, checks every identity of these appendices with the cocoa Python, 23 checks that each print one PASS or FAIL line with the measured deviation: the two-bin, three-bin, even and uneven tridiagonal inverses and the projection's null direction, semidefiniteness and rank; the Pythagorean split and the Schur complement; the neighbor-prediction leftover and, by Monte Carlo, the jump variance with the worked example of Appendix A.2; the continuity equation against Eq. (14), the conserved deviations and the neighbor rms values; and for the memory model the annihilation of U with rank $n-2$, the $1/W^2$ Woodbury limit, the null-space identity, the three-bin curvature penalty, the vanishing for one and two bins, the kriging predictor, the end-to-end correlation, the straight-line fits of Appendix C.4 and the LRG3 magnitude-limit rate. The theory figures are drawn from data written by `plots/make_theory_data.py`, from the same flat ΛCDM growth factor and fiducial amplitudes.

M Implementation and the Fisher construction

M.1 Model A in the likelihood

Model A runs in the project likelihood (`likelihood/desi_cl.py`) when its option `higher_order_bias` is 1; the example yamls `EXAMPLE_EVALUATE_MODEL_A1–A4.yaml` set it up on the four layouts, in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2. Every model A parameter is per bin in the likelihood; the grouping, one value per group or a value plus a tilt, is a statement of the parameter yamls, generated by `scripts/make_desi_params.py --model-a`.

- **Quadratic and tidal bias.** The likelihood computes $b_{2,i}$ and $b_{s^2,i}$ from $b_{1,i}$ with Eq. (5) and passes them to CosmoLike through the setter that takes the tidal bias per bin.

- **Counterterm.** CosmoLike's k^2 slot is $2\,b_1 b_K\,k^2 P_{\rm mm}$, with b_K its per-bin derivative-bias coefficient and k in units of H_0/c. The likelihood sets $b_K = \alpha_0/[2\,b_1(c/H_0)^2]$, with $c/H_0 = 2997.92\,h^{-1}$Mpc. The counterterm therefore multiplies the nonlinear matter spectrum, CosmoLike's convention, where DESI's definition multiplies the linear one (subsection "The draft's data vector against CosmoLike" of `notes/higher_order_bias`).

- **Stochastic term.** A constant three-dimensional power projects exactly, without the

Limber approximation, to an ℓ-independent spectrum,

$$C_\ell^{ij}\big|_{\rm stoch} = {\rm SN}_0\, P_{ij}\, K_{ij}, \qquad K_{ij} = \int dz\, n_i(z)\, n_j(z)\, \frac{dz/d\chi}{\chi^2}, \tag{63}$$

with n_i the normalized Fingers-of-God-convolved redshift distribution of bin i, χ the comoving distance, and P_{ij} the Poisson power of the pair. For one bin, $P_{ii} = 1/\bar{n}_i$, fixed by the bin's surface density so that the projection with the raw distribution gives the angular shot noise. A cross pair of two bins of one tracer sample takes the mean of their two powers; a pair of two samples takes none, because different galaxies share no stochastic term.

- **Projection.** The low-rank non-Limber engine projects each bin's full one-loop $P_{gg,i}$, counterterm included, and the adjacent cross pairs carry the bilinear one-loop pair spectrum (subsection "Gaps against model A" of `notes/higher_order_bias`). Appendix P shows why the counterterm requires this engine.

M.2 The derivatives

The Fisher matrix is built from the derivatives of the data vector m (the band powers) with respect to every parameter. The notebook `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb` builds them in three groups.

Stencil derivatives. The seven cosmological parameters ($10^9 A_s$, the spectral index n_s, the Hubble constant H_0, the baryon density Ω_b, Ω_m, w_0, w_a), the six group amplitudes B_g and the velocity dispersions come from a five-point stencil of the model A likelihood.

Per-bin operator columns. The four columns $\partial m/\partial b_{1,i}$, $\partial m/\partial b_{2,i}$, $\partial m/\partial b_{s^2,i}$ and $\partial m/\partial \alpha_{0,i}$ of every bin, each at fixed values of the other three, come from central differences through the likelihood's bias setter, with steps of 1% of b_1, 0.02 in b_2 and b_{s^2}, and $5\,h^{-2}\,{\rm Mpc}^2$ in α_0. A wrapper shifts one bias for a set of bins: one color of a greedy coloring of the cross-pair graph, in which two bins joined by a measured cross pair never share a color. Each spectrum then involves at most one shifted bin, so one pair of evaluations gives the columns of every bin of that color. Every layout takes four colors, so each operator costs eight evaluations instead of two per bin.

Stochastic columns. The ${\rm SN}_0$ columns are the exact templates $P_{ij}K_{ij}$ of Eq. (63).

Model A ties b_2 and b_{s^2} to b_1, so the derivative with respect to a bin's linear bias is the chain-rule sum

$$\left.\frac{dm}{db_{1,i}}\right|_{\rm A} = \frac{\partial m}{\partial b_{1,i}} + b_2^{\rm L\prime}(b_{1,i})\,\frac{\partial m}{\partial b_{2,i}} - \frac{4}{7}\,\frac{\partial m}{\partial b_{s^2,i}}, \tag{64}$$

with $b_2^{\rm L\prime}$ the slope of the Lazeyras fit. Summed over a group with weights $1/D(\bar{z}_i)$, these columns must reproduce the stencil derivative of the amplitude B_g (Appendix O).

M.3 The variants and the Fisher matrices

The parameters are, in order: the cosmological parameters and the velocity dispersions (with the Gaussian priors of the linear-bias run), the variant's higher-order parameters, and one amplitude $\beta_i = b_{1,i}\,D(\bar{z}_i)$ per bin, fiducial $\beta_i = B_g$ (Eq. (9)). The column of β_i is Eq. (64)

divided by $D(\bar{z}_i)$. The higher-order columns of group g are sums over its bins,

$$\frac{\partial m}{\partial \Delta_{2,g}} = \sum_{i \in g} \frac{\partial m}{\partial b_{2,i}}, \qquad \frac{\partial m}{\partial t_{2,g}} = \sum_{i \in g} (\bar{z}_i - \bar{z}_g) \, \frac{\partial m}{\partial b_{2,i}}, \tag{65}$$

and likewise for Δ_s and α_0, with $t_{2,g}$ the tilt of Δ_2 in group g; the SN_0 column sums the group's stochastic templates. The exact-calibration configuration of Appendix Q has no higher-order column, model A baseline has the offset columns with their Gaussian priors, and model A pessimistic adds the tilt columns, with flat priors, for every group of more than one bin.

The data Fisher matrix is $F_{\mathrm{data}} = \mathsf{D}^{\mathsf{T}} \mathsf{C}_d^{-1} \mathsf{D}$ on the kept entries, with D the matrix of the columns above and C_d the band-power covariance: Gaussian, with linear bias $B_g/D(\bar{z}_i)$, computed for 14,000 square degrees, its inverse rescaled to the survey's 12,300. The Gaussian priors add to its diagonal. The higher-order parameters are marginalized like the cosmology, so the ladder code of the linear-bias run acts unchanged on the amplitude block. Every Fisher matrix is inverted through a pseudo-inverse (Appendix O).

M.4 The ladder, the response and the truths

- **Linear-bias priors.** Growth-tied sums the amplitude columns of each group into one; the baseline and pessimistic priors add the projected precision of the residual prior, Eq. (21), to the amplitude block (Appendices K and J); free adds nothing.
- **Response.** A truth that moves the data by δm moves the best fit by $\delta p = F^{-1} \mathsf{D}^{\mathsf{T}} \mathsf{C}_d^{-1} \delta m$, to first order (Appendix F.1).
- **Known drifts.** The two drift laws of Appendix F.2. Under model A a drift δ_i of the linear bias moves b_2 and b_{s^2} as well: $\delta m = \sum_i (\partial m/\partial \beta_i)_{\mathrm{A}} \, \beta_i \, \delta_i$, with the model A column of Eq. (64).
- **Random truths and $\sqrt{\langle \chi^2 \rangle / n_{\mathrm{dof}}}$.** The 2000 draws per scatter family of the linear-bias run, with its seed and order (Appendix F.3), and the statistic of Eq. (6) (Appendices F.4 and F.5).
- **Calibration drift.** One deterministic truth that moves only the offsets: $\delta m = \sum_i [(\partial m/\partial b_{2,i}) \, \Delta_{2,i} + (\partial m/\partial b_{s^2,i}) \, \Delta_{s,i}]$, with $\Delta_{2,i} = (\bar{z}_i - \bar{z}_g)/(\bar{z}_{\max} - \bar{z}_{\min})$ and $\Delta_{s,i} = \Delta_{2,i}/2$, where $\bar{z}_{\max}$ and $\bar{z}_{\min}$ are the largest and smallest mean redshifts of the group's bins. The drift changes by 1 in Δ_2 and 0.5 in Δ_s from the group's first bin to its last and has zero mean over the group's bins; in a one-bin group, where the ratio is undefined, $\Delta_{2,i}$ is set to zero.

Every figure lives in its own folder under `plots/`, named after the figure, with its data (`data.json`), the Python script that draws it, and the PDF the note includes; each figure redraws from its folder alone.

N The known drifts

Figure 10 draws the shift of each known drift alone under model A baseline. The free prior absorbs both drifts exactly. The growth-tied prior stays excluded: the conserved-population

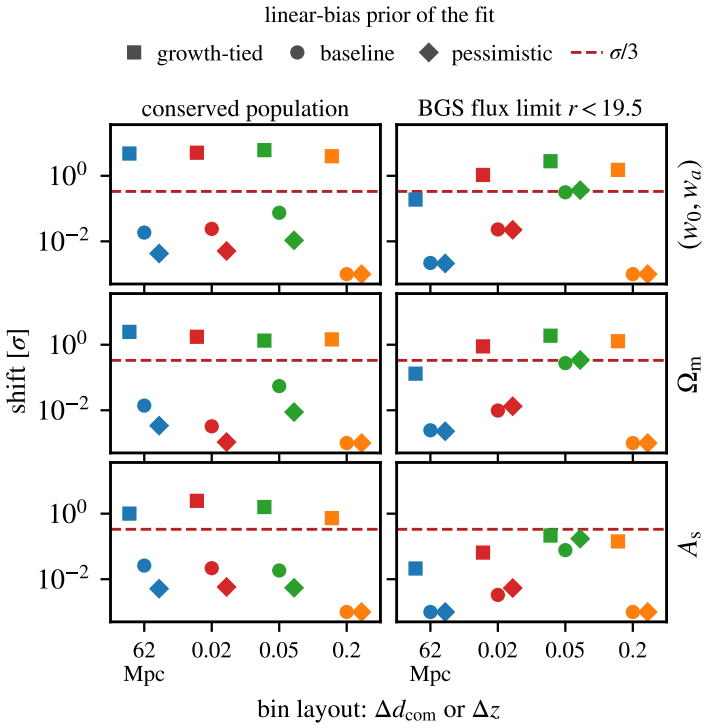


Figure 10: The shift of a model A baseline fit under each known drift alone: the conserved-population drift (left) and the BGS $r < 19.5$ flux limit (right); (w_0, w_a) in units of the fit's ellipse, Ω_m and A_s in their own σ. Markers: the linear-bias prior of the fit; colors: the layouts. Dashed line: $\sigma/3$. Points on the bottom axis: values below 10^{-3}, among them the exact zeros of the baseline and pessimistic priors on $\Delta z = 0.2$. Omitted: the free prior, which absorbs both drifts exactly, and model A pessimistic. $k_{\max} = 0.075\,h/\mathrm{Mpc}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

drift moves its (w_0, w_a) by several σ, and by 1.2–1.8σ under model A pessimistic. The baseline and pessimistic priors absorb both drifts below $\sigma/3$ in every score but one case: the BGS flux limit on $\Delta z = 0.05$, a layout that resolves BGS into only six bins. There the baseline prior sits just below $\sigma/3$, and the pessimistic prior no longer absorbs the drift: it lands just above $\sigma/3$ in (w_0, w_a) and in Ω_m. This residual drift is what lifts the pessimistic prior's η on that layout to its maximum of 1.05 (Appendix F.4). Under model A pessimistic both priors stay below $\sigma/10$. The full numbers, every prior, both variants and both truths, are in the `data.json` of each figure and in the printed tables of `EXAMPLE_EVALUATE_FISHER_MODEL_A.ipynb`.

O Numerical checks

- **The chain rule.** For every group, the per-bin columns of Eq. (64), summed with the weights $1/D(\bar{z}_i)$, reproduce the stencil derivative of the group amplitude B_g to 9.8×10^{-8}, 1.2×10^{-7}, 4.0×10^{-8} and 1.3×10^{-8} of the stencil column's largest entry (layout order): the operator columns through the bias setter and the full likelihood through the stencil

agree.

- **The matched-prior identity.** When a fit's prior equals the truth's scatter, the identity $\mathsf{P\Sigma P} = \mathsf{P}$ of Appendix F.4 makes the scatter and noise terms of $\langle\chi^2_w\rangle$ sum to exactly 2. Over every layout and variant, $\max|\langle\chi^2_w\rangle - \Delta^2_{\rm drift} - 2| = 9.4 \times 10^{-10}$.

- **Exact recovery.** The free prior must absorb any truth of the linear bias. Its largest mean (w_0, w_a) shift over the random truths is 1.9×10^{-11}.

- **One direction without information.** On $\Delta d_{\rm com} = 62$ Mpc at this cut only 7 of the 18 BGS bins keep any band. Under model A pessimistic on the free prior, the flat-prior BGS tilts of Δ_2, Δ_s and α_0 then leave one combination of parameters with no information, and a plain matrix inverse amplified round-off along it into a spurious 2σ shift. The notebook therefore scales every Fisher matrix to unit diagonal, diagonalizes it, and drops the directions whose eigenvalue is below 10^{-12} of the largest. Exactly one of the 48 systems (four layouts, four linear-bias priors, and three higher-order configurations: model A baseline, model A pessimistic and the exact calibrations of Appendix Q) drops one direction, whose largest component on (w_0, w_a), Ω_m and A_s is 6.3×10^{-12}: a pure nuisance combination.

- **The α_0 step.** At the negative stencil point, $\alpha_0 = -5\,h^{-2}\,{\rm Mpc}^2$, the factor $b_1^2 + \alpha_0 k^2$ of the linear and counterterm pieces turns negative above $k \approx 0.6\,h/{\rm Mpc}$, where the low-rank engine floors $\sqrt{P_{gg}}$. On $\Delta z = 0.05$, a step of 5 against a step of 1 changes the α_0 columns by at most 3.5×10^{-4} of each column's largest entry.

- **Reproducibility.** A recomputation of the $\Delta z = 0.05$ derivatives is bitwise identical to the cache.

- **The loop templates.** With $b_3 = 0$, CosmoLike's C-FAST-PT loop templates and Python FAST-PT agree on the $\Delta d_{\rm com} = 62$ Mpc vector, pairs included, to a $\Delta\chi^2$ far below one (subsection "Gaps against model A" of `notes/higher_order_bias`).

P The counterterm requires the low-rank engine

CosmoLike has two non-Limber engines for C_ℓ^{gg}. FKEM, the FFTLog method of Fang et al. [8], projects the linear term exactly and moves everything nonlinear, the counterterm included, through its Limber difference, so its one-loop content is Limber. The low-rank engine factorizes each bin's full one-loop P_{gg}, less the white-noise constant of the loops (which Limber projects exactly), and projects it without the Limber approximation.

For the counterterm the difference is large. On the $\Delta z = 0.05$ layout the low-rank α_0 response, $\partial C_\ell/\partial\alpha_0$, is 7.3 times the Limber one at $\ell = 20$, 1.8 times at $\ell = 50$, 1.26 times at $\ell = 100$ and about 1.05 times at $\ell = 200$–400 (median over the kept auto spectra). The reason is the radial modes of thin bins. Limber evaluates the power at the transverse wavenumber $k = (\ell + 1/2)/\chi$ only. A thin bin with sharp edges also passes radial modes of higher k, and the k^2 weight of the counterterm amplifies them; at low ℓ, where the transverse wavenumber is small, they dominate the response. The low-rank response is converged: raising `nonlimber_accuracyboost` from 32 to 128, the rank from 5 to 10, or the overall accuracy boost from 1 to 2 changes it by a $\Delta\chi^2$ far below one. For a one-sigma α_0 the two engines' responses

differ by tens of units of χ^2, while at $\alpha_0 = 0$ they agree on the full vector to a few units. The forecast uses the low-rank engine.

Q Comparison with linear bias and with exact calibrations

Neither comparison in this appendix is a result of this note. The linear-bias run uses a model that `notes/linear_bias_kmax` shows is already biased at this cut, and exact calibrations assume that b_2 and b_{s^2} are known exactly. Both sets of FoMs are in `plots/baseline_ladder/data.json` and in the notebook's printed ladder table.

Linear bias. Model A baseline keeps about 88, 93, 74 and 50% of the linear-bias run's FoM(w_0, w_a) on the baseline prior, and model A pessimistic about 82, 81, 55 and 30% on the pessimistic prior (layouts in the order $\Delta d_{\rm com} = 62$ Mpc, $\Delta z = 0.02$, 0.05, 0.2). The loss grows as the bins widen. A likely reason: on the wide layouts a group holds few bins, so one offset per group is close to one offset per bin, and the new parameters compete with each bin's linear bias for the same band powers.

Exact calibrations. With the calibrations taken as exact, the FoM exceeds that of the linear-bias run under every prior and on every layout: on the baseline prior about 1.4 times on $\Delta d_{\rm com} = 62$ Mpc and 2.0 times on $\Delta z = 0.05$. The loop terms are then not free: with $b_{2,i}$ and $b_{s^2,i}$ fixed by the calibrations at $b_{1,i}$, and $\alpha_0 = {\rm SN}_0 = 0$, the one-loop correction is a known function of $b_{1,i}$ and of the matter power spectrum. The loop terms grow as the fourth power of the growth factor and the linear term as b_1^2 times its square, so the size of the correction relative to the linear term carries information on the amplitude of the matter fluctuations that a linear-bias model does not have. That information is only as good as the calibrations: with exact calibrations the calibration drift shifts (w_0, w_a) by 0.8–1.9σ under the growth-tied prior and by 0.06–0.48σ under the other three. The offsets of model A baseline, with widths set by the known scatter of the calibrations, and its α_0 and ${\rm SN}_0$ remove this information, and model A baseline sits below the linear-bias run under every prior and on every layout.

R Limitations

- **Linear response.** The shifts and $\sqrt{\langle\chi^2\rangle/n_{\rm dof}}$ are first order in the truth's departure from the fiducial. The dropped second-order term is a few percent of the shift for 1–3% residuals and perhaps 10–20% for the 22% BGS drift (Appendix F.3).
- **Theory truths, not mocks.** The project is a theory paper. The truths are the derived drifts, Gaussian residuals of the two stated sizes, and one linear drift of the calibrations. Residuals rougher than the pessimistic prior, and calibrations that drift nonlinearly within a group, are untested.
- **Real-space loops.** Model A adds the real-space one-loop terms to linear Kaiser distortions.

S Open issues

- **One-loop terms in the covariance.** The covariance is Gaussian with linear bias $B_g/D(\bar{z}_i)$; it leaves out the one-loop terms of model A, which make up about 2% of the kept spectra (median 1.6–2.3% over the layouts).

- **The scale cut was not rebuilt on the one-loop model.** Applied to the α_0 template $\partial C_\ell/\partial\alpha_0$, the UV fraction of Eq. (1) is a median 0.076 and a maximum 0.14 on the kept bands, all above 0.05, against a median 0.017 and a maximum 0.052 for the fiducial model A vector. Part of what the forecast learns about α_0 therefore comes from scales the cut is meant to exclude.

- **Negative α_0 in chains.** The linear and counterterm pieces combine to $(b_1^2+\alpha_0 k^2)\,P_{\rm mm}$, negative above $k \approx b_1/\sqrt{|\alpha_0|}$. At $\alpha_0 = -88.9\,h^{-2}\,\mathrm{Mpc}^2$, one prior width below zero, the spectrum the low-rank engine factorizes turns negative above $k \approx 0.14\,h/\mathrm{Mpc}$, the engine floors 47% of its grid, and the vector curves in α_0, which is harmless for the Fisher forecast at $\alpha_0 = 0$, but not for chains that explore negative α_0. The two remedies, a loop taper above the scale where one-loop perturbation theory is trusted and a positivity prior on P_{gg} (subsection "The white-noise limit of the renormalized loops" of `notes/higher_order_bias`), are not implemented.

- **The b_3 template is broken.** CosmoLike's third-order bias template is a fixed table that is wrong even at the fiducial; model A sets $b_3 = 0$, as in DESI's baseline, where the template drops out (`notes/higher_order_bias`).

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